

BFS Traversal

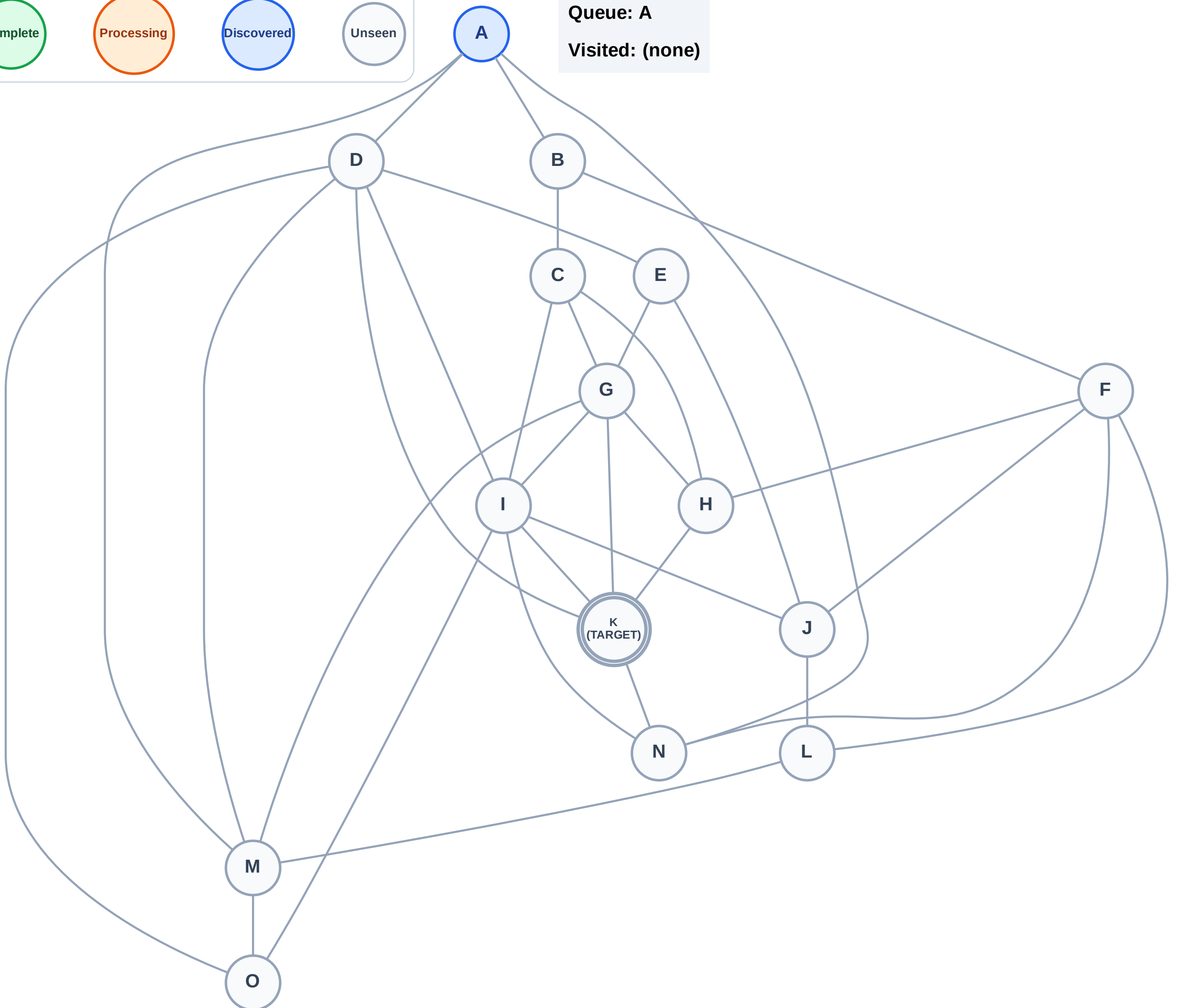
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

Complete

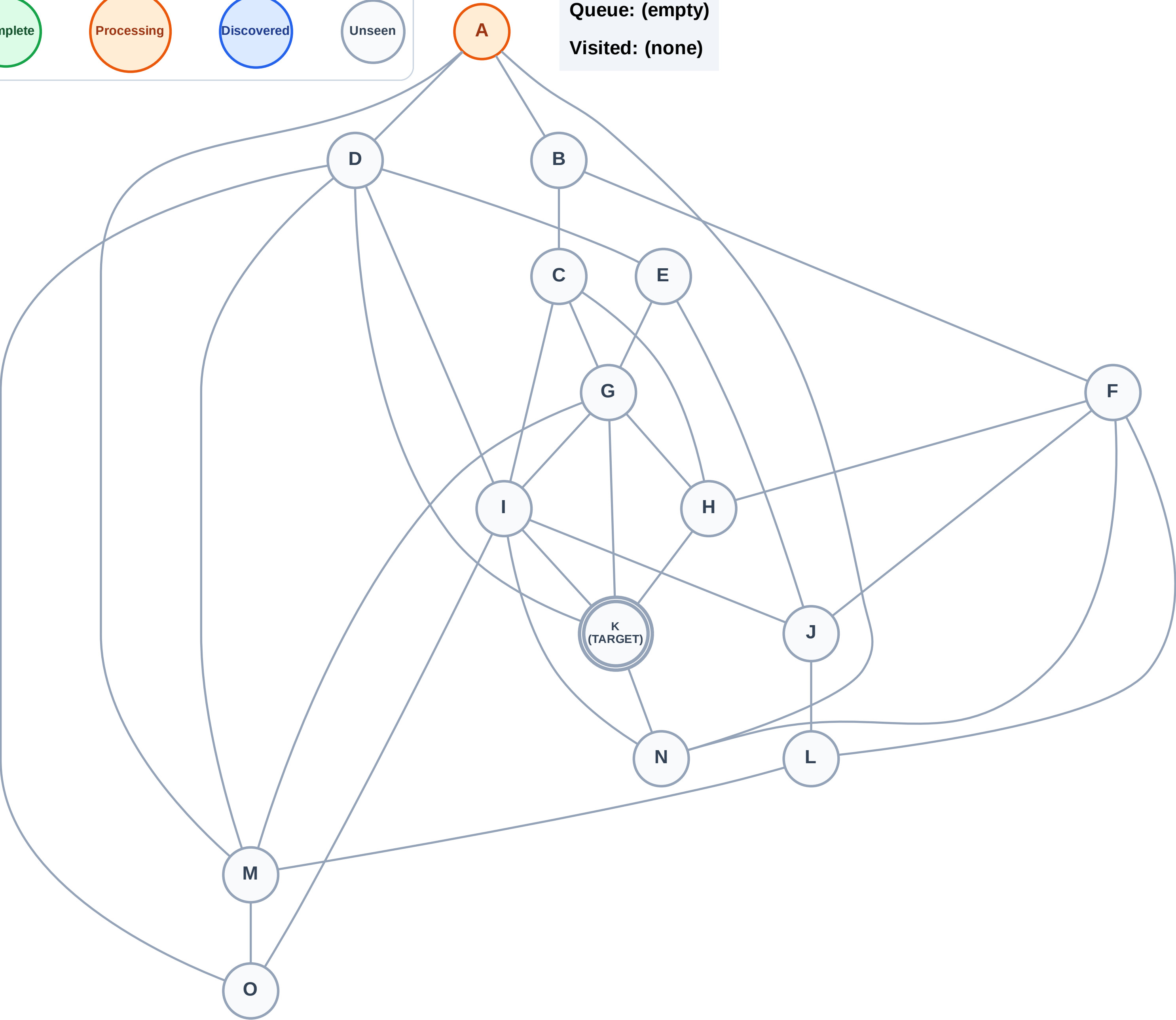
Processing

Discovered

Unseen

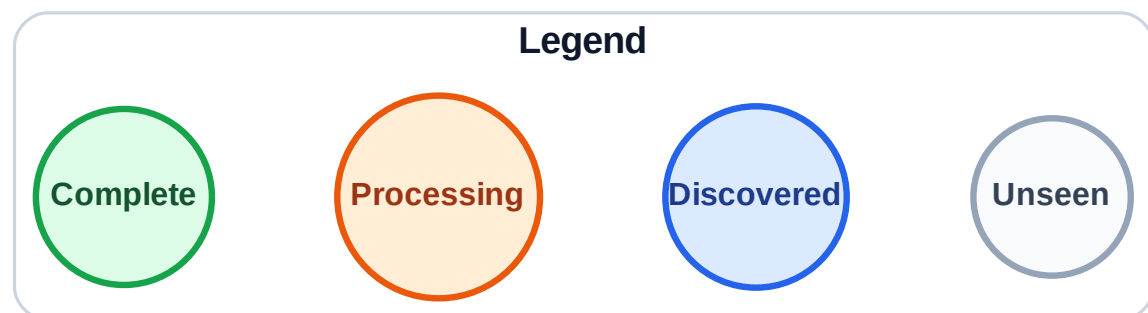
Queue: (empty)

Visited: (none)

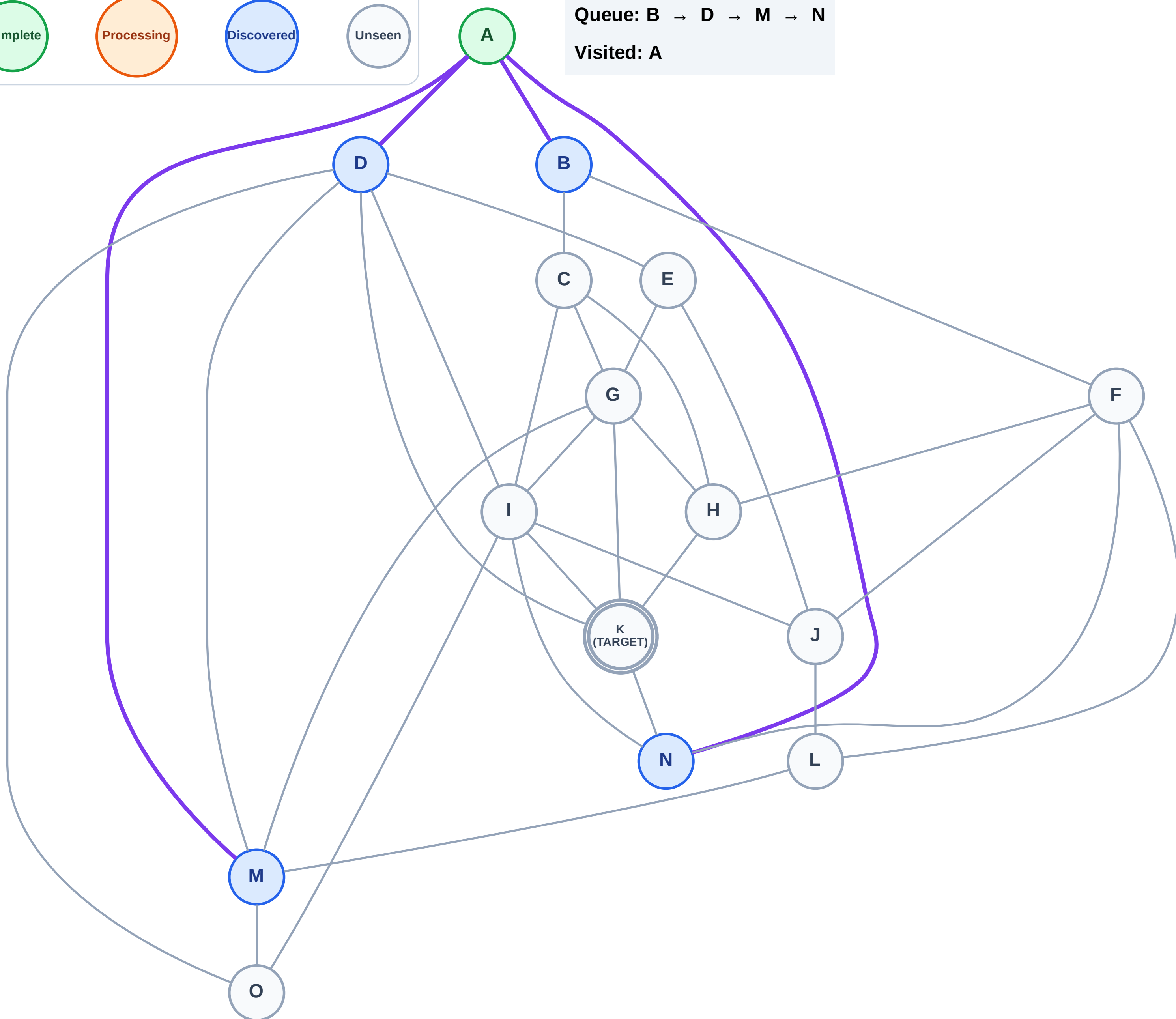


BFS Traversal

Step 3: Discover B, D, M, N from A



Queue: B → D → M → N
Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

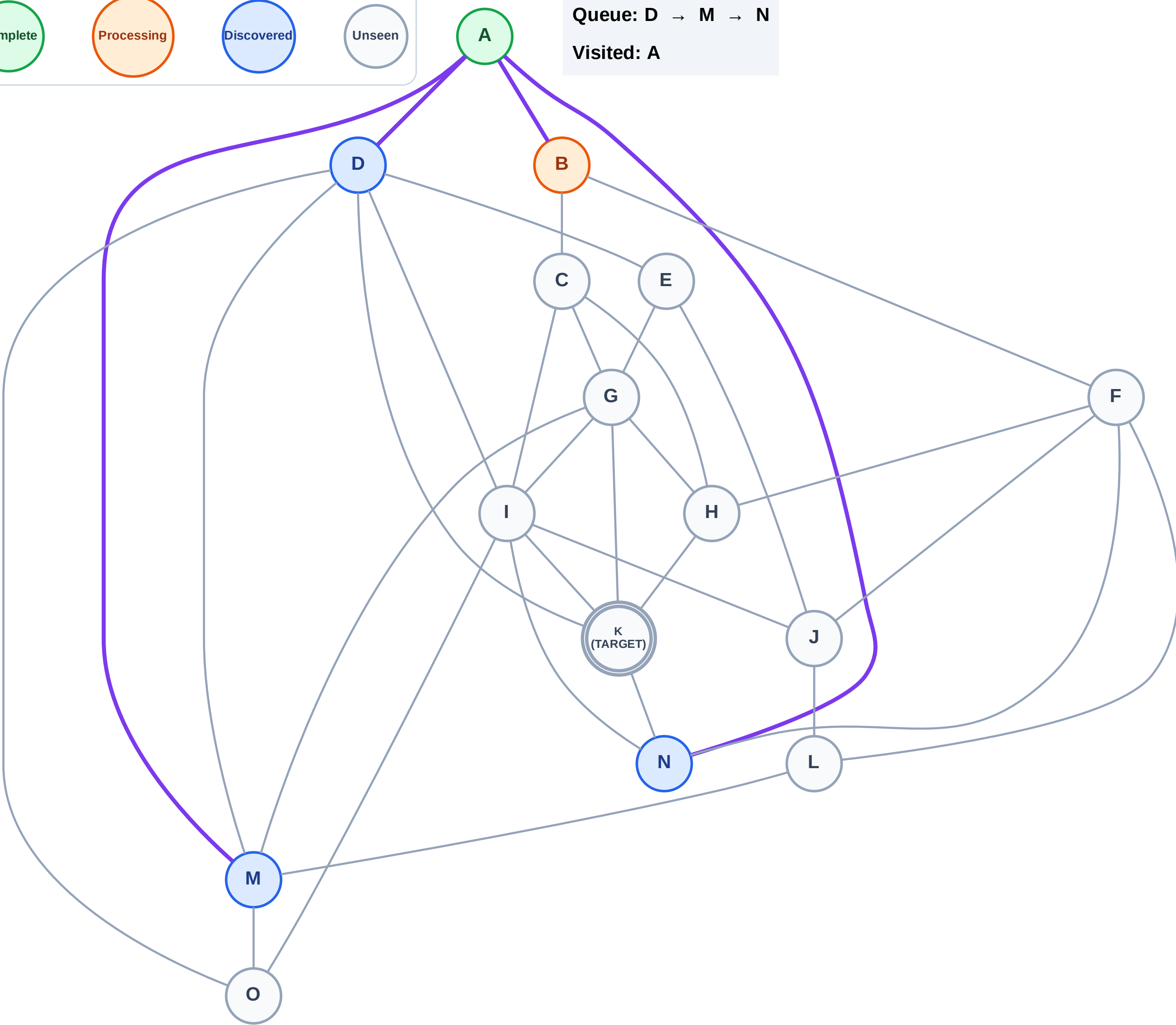
Processing

Discovered

Unseen

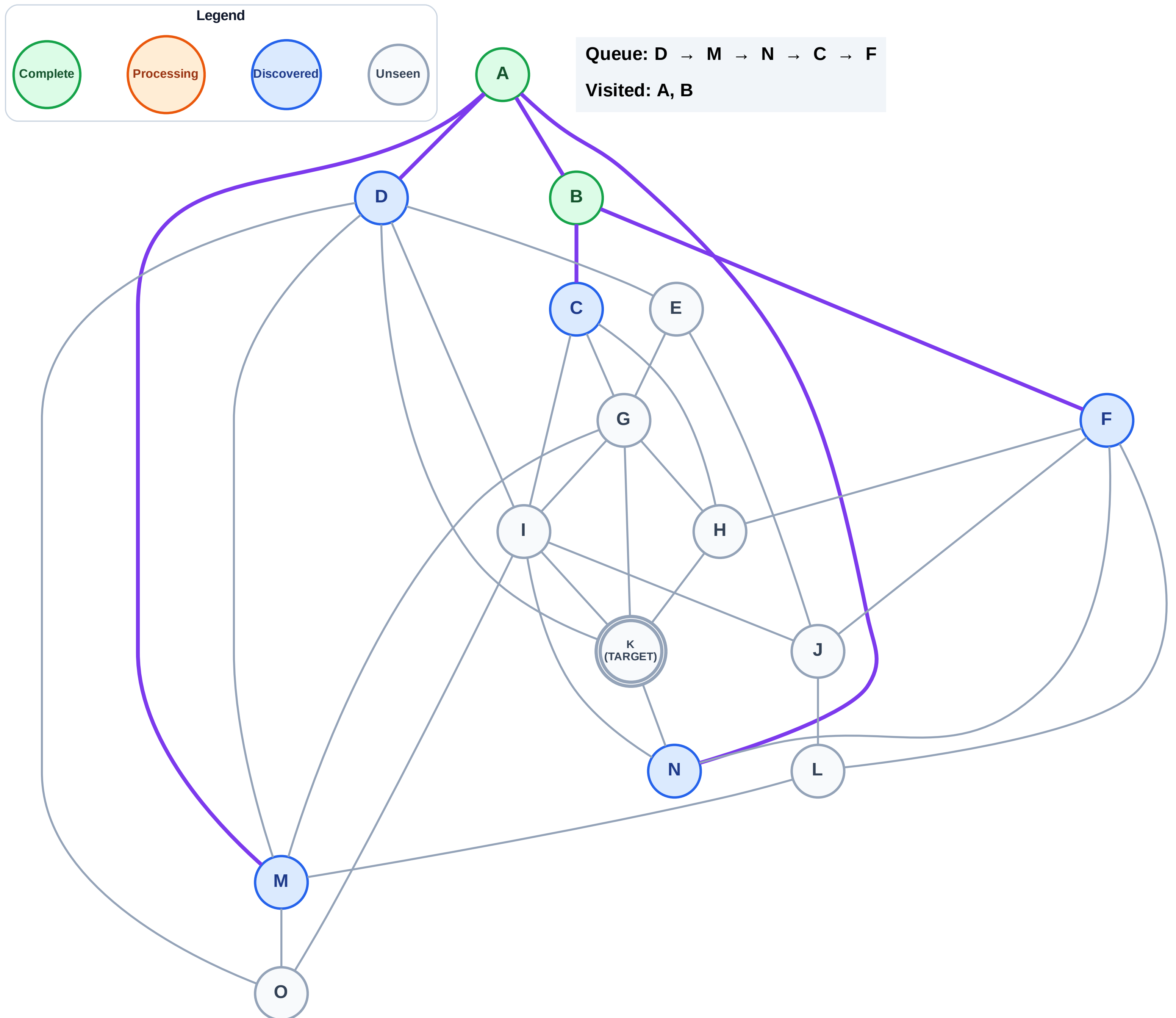
Queue: D → M → N

Visited: A



Step 5: Discover C, F from B

Queue: D → M → N → C → F
Visited: A, B



BFS Traversal

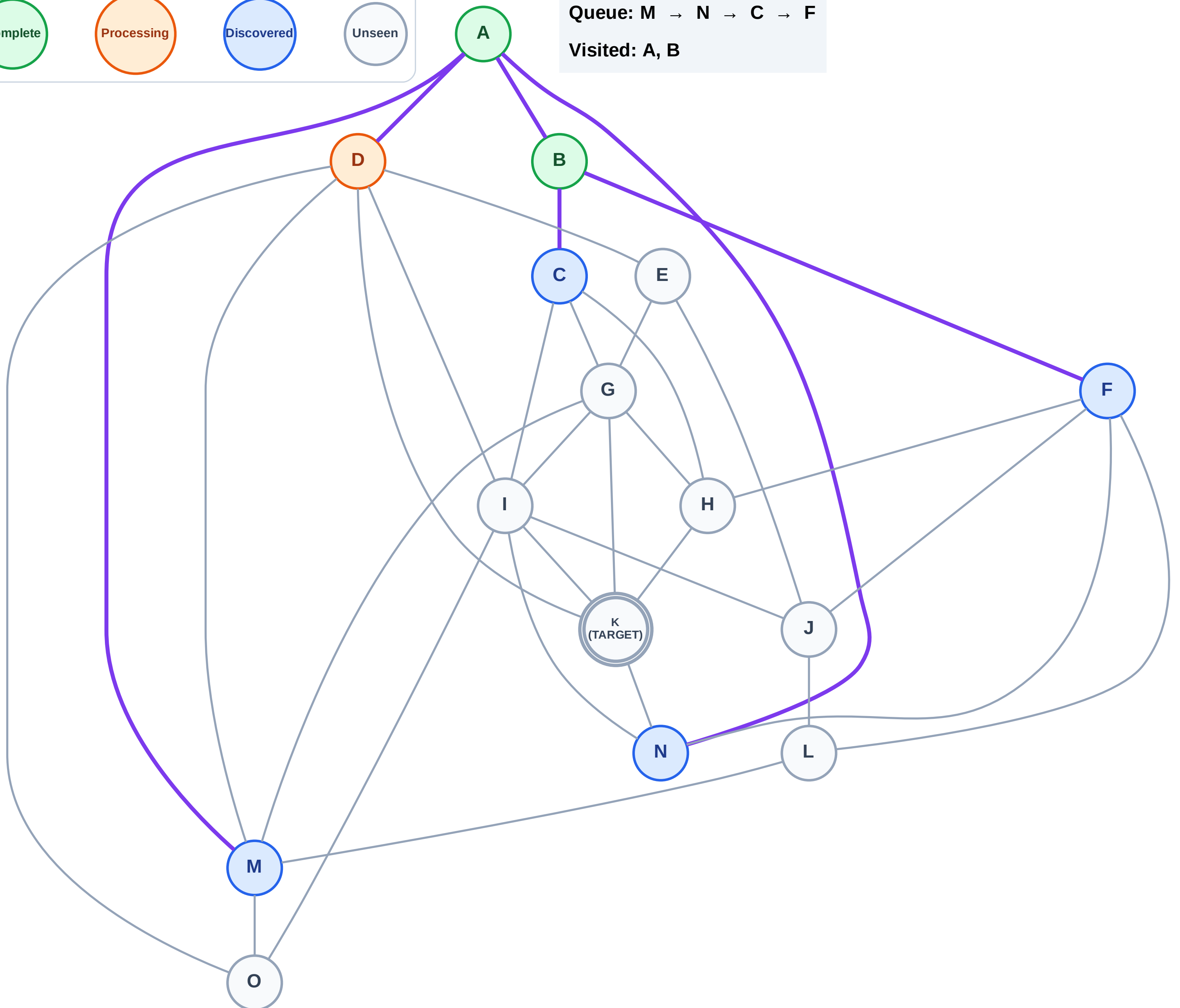
Step 6: Process node D

Legend



Queue: M → N → C → F

Visited: A, B



BFS Traversal

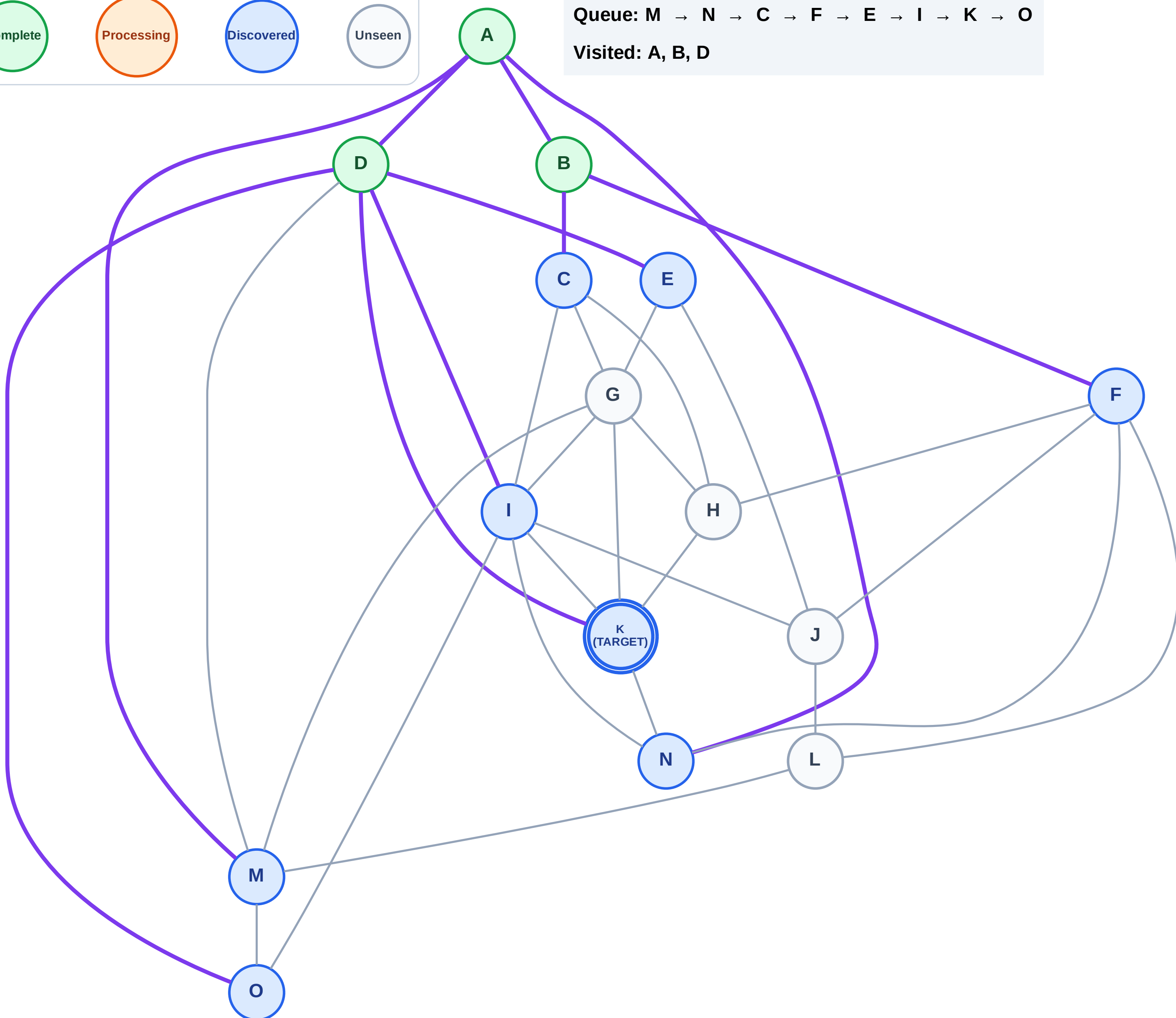
Step 7: Discover E, I, K, O from D

Legend



Queue: M → N → C → F → E → I → K → O

Visited: A, B, D



BFS Traversal

Step 8: Process node M

Legend

Complete

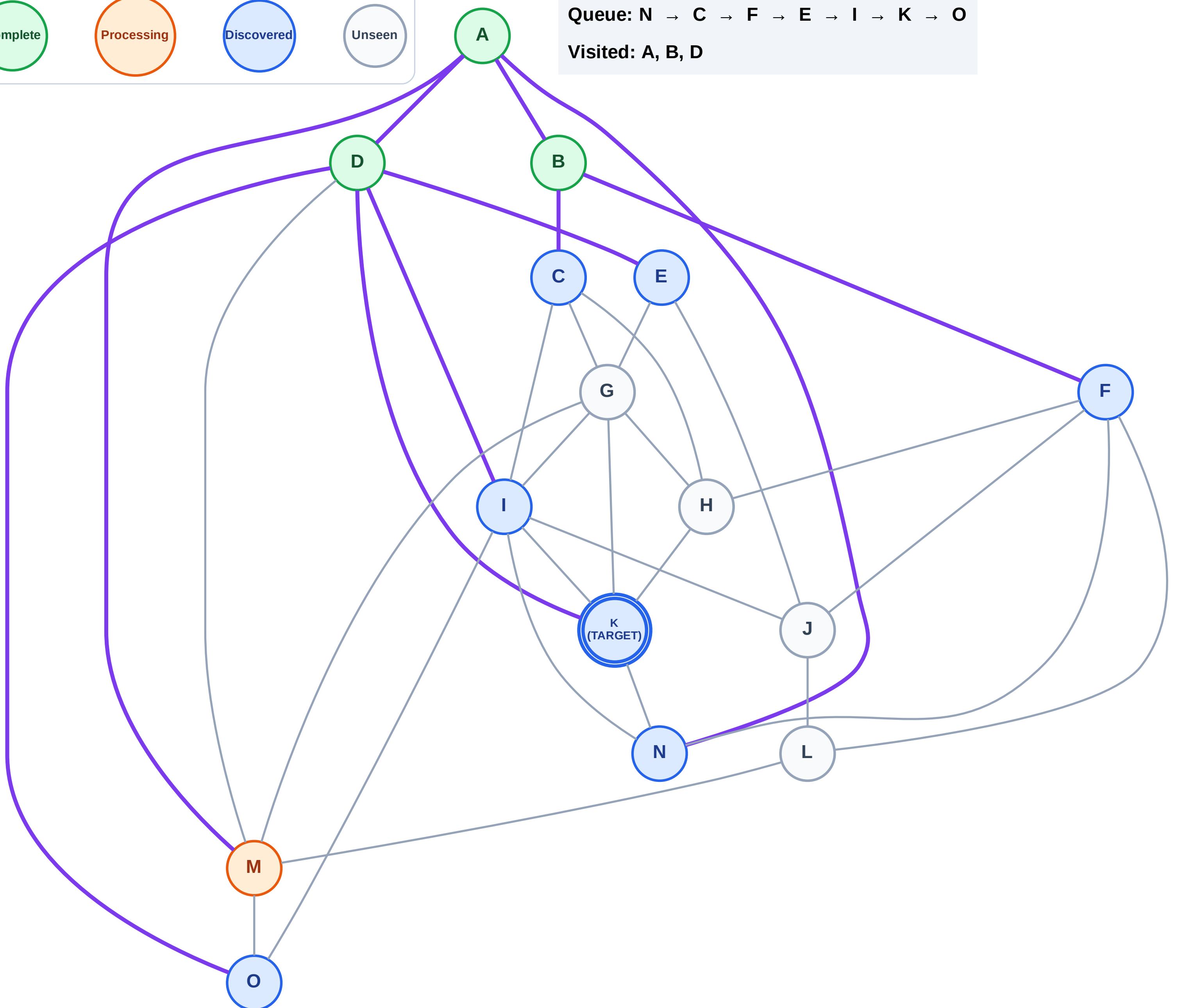
Processing

Discovered

Unseen

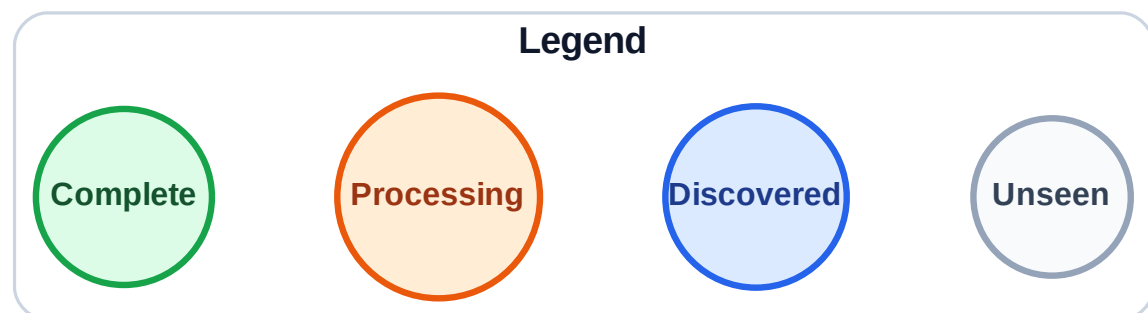
Queue: N → C → F → E → I → K → O

Visited: A, B, D

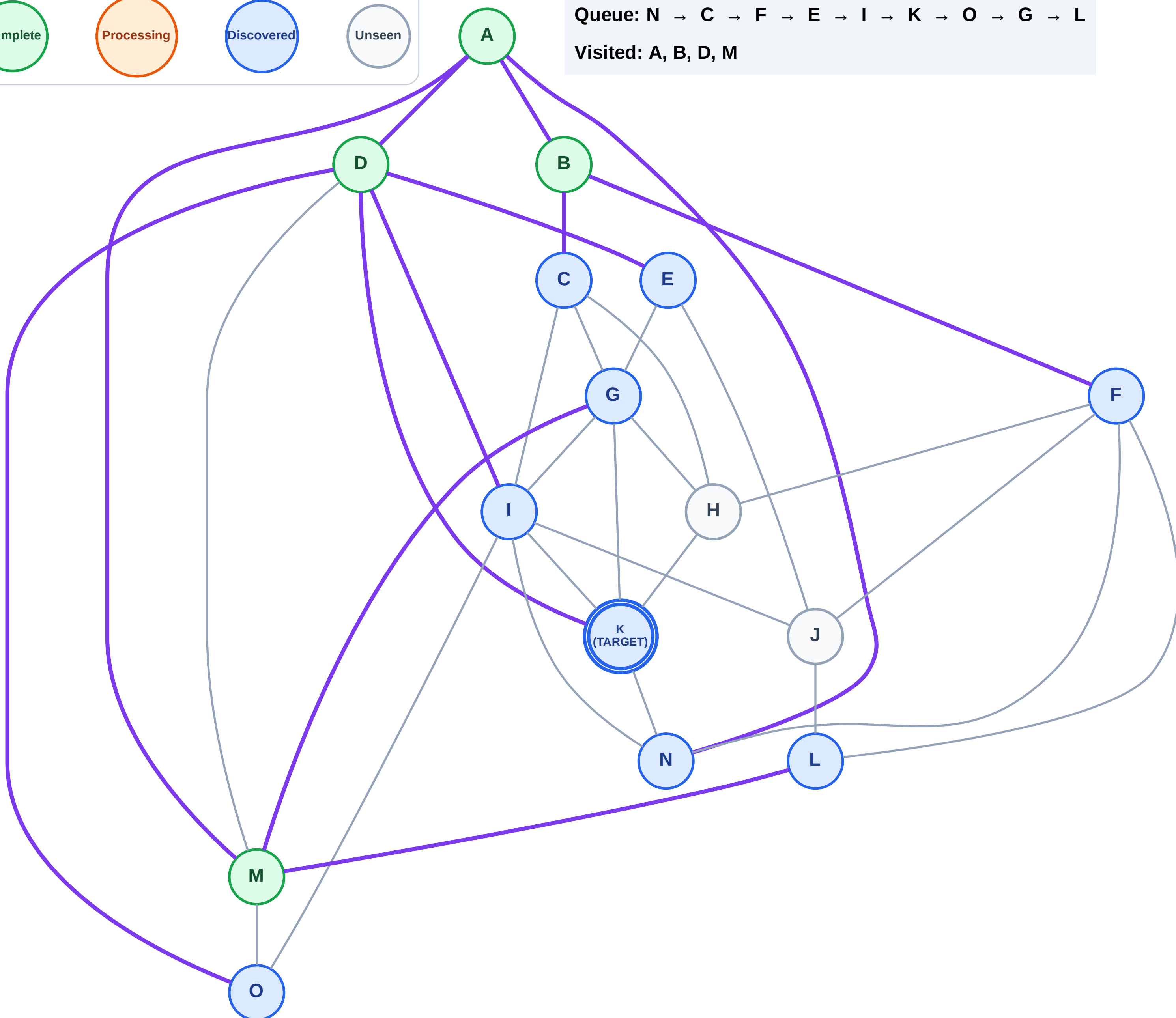


BFS Traversal

Step 9: Discover G, L from M



Queue: N → C → F → E → I → K → O → G → L
Visited: A, B, D, M



BFS Traversal

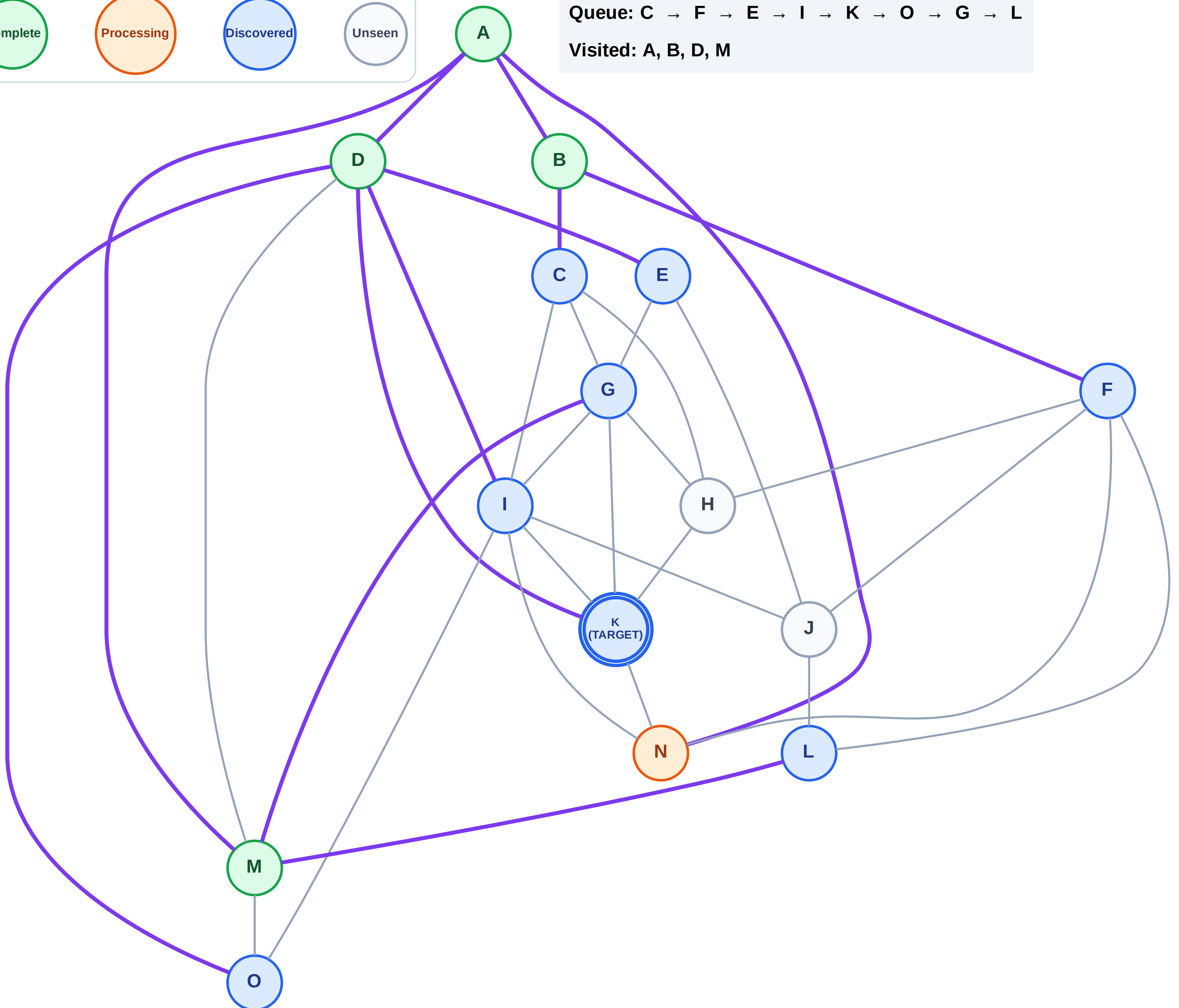
Step 10: Process node N

Legend



Queue: C → F → E → I → K → O → G → L

Visited: A, B, D, M



BFS Traversal

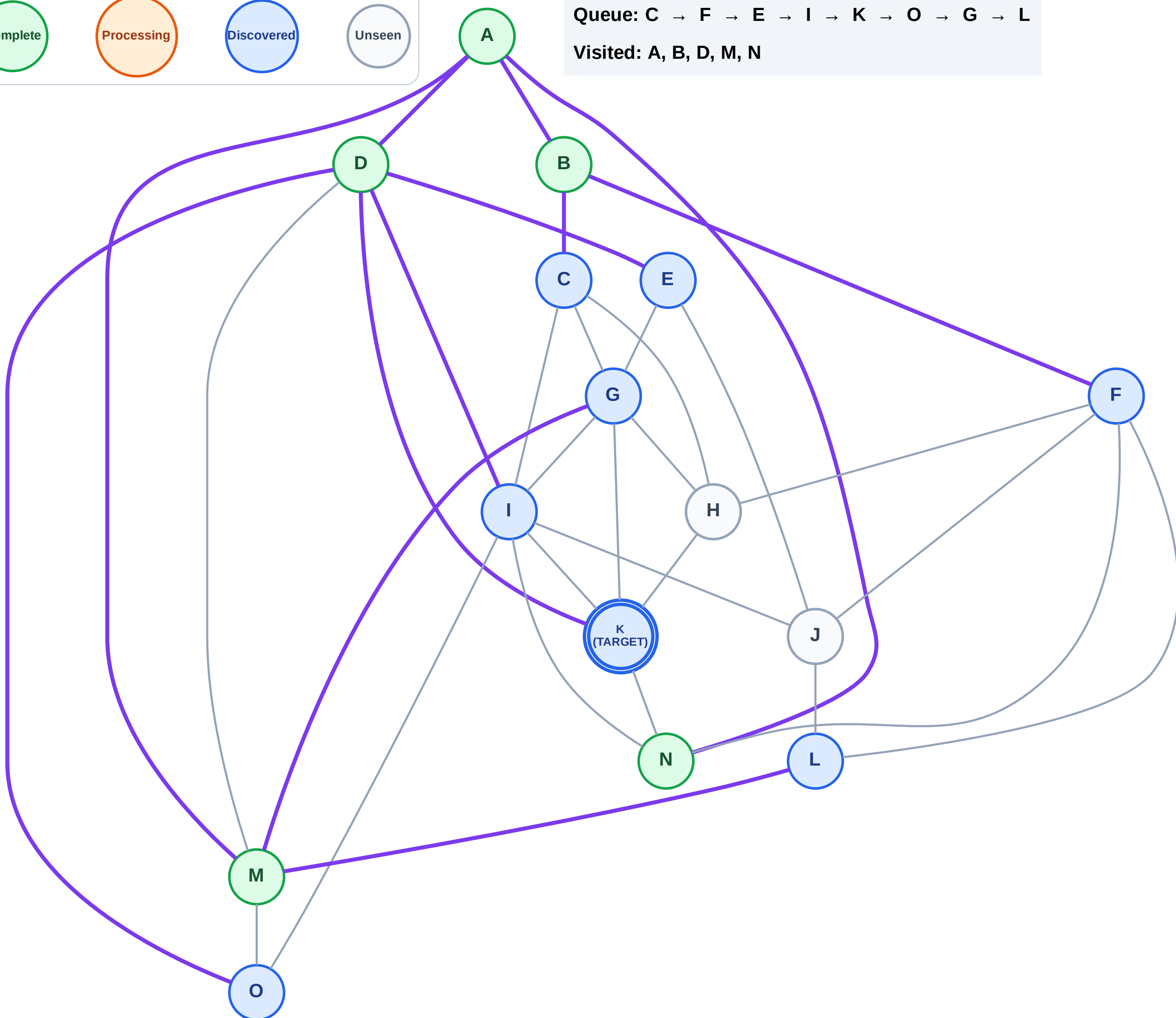
Step 11: No new neighbors from N

Legend



Queue: C → F → E → I → K → O → G → L

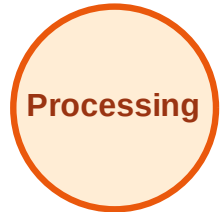
Visited: A, B, D, M, N



BFS Traversal

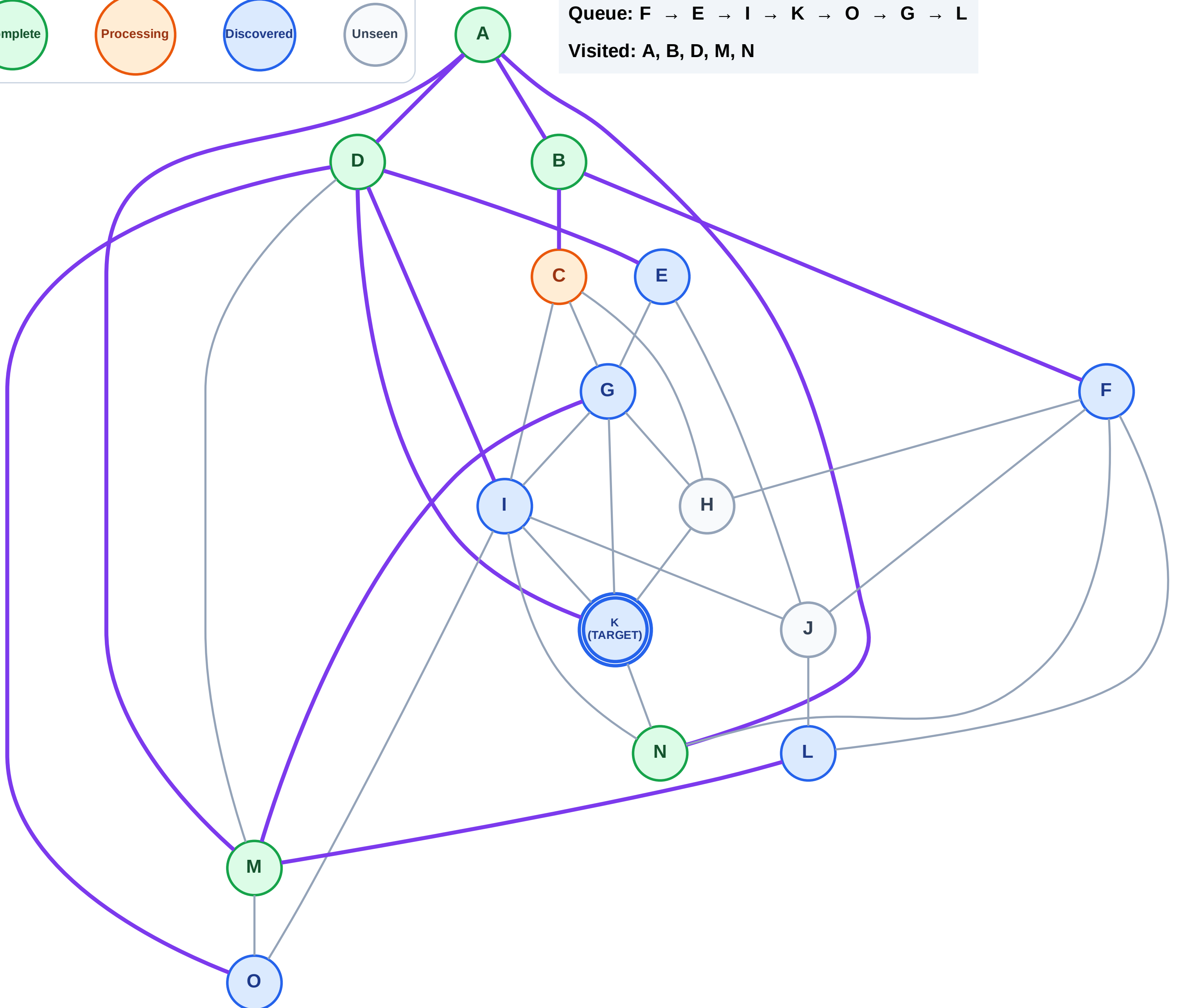
Step 12: Process node C

Legend



Queue: F → E → I → K → O → G → L

Visited: A, B, D, M, N



BFS Traversal

Step 13: Discover H from C

Legend

Complete

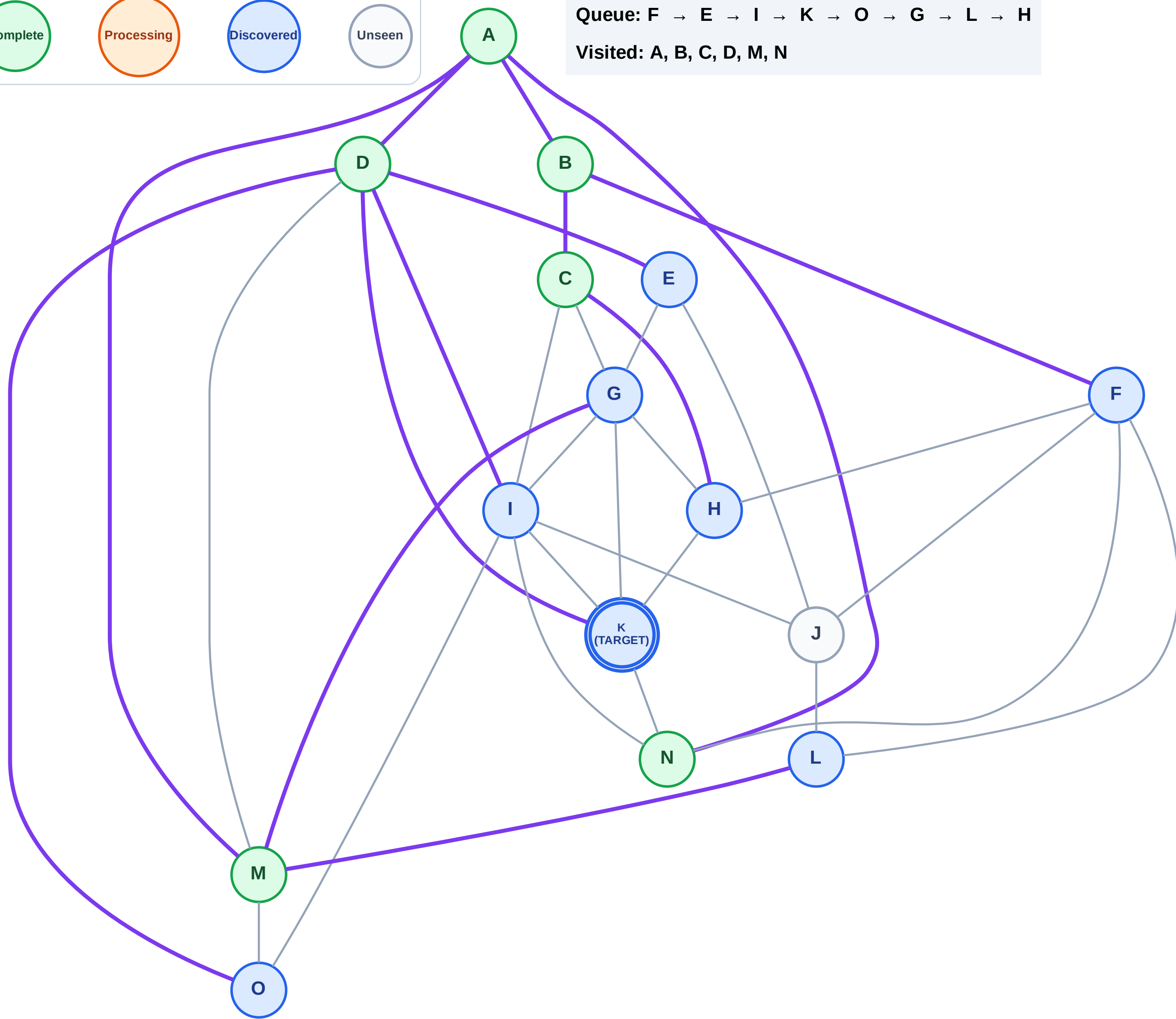
Processing

Discovered

Unseen

Queue: F → E → I → K → O → G → L → H

Visited: A, B, C, D, M, N



BFS Traversal

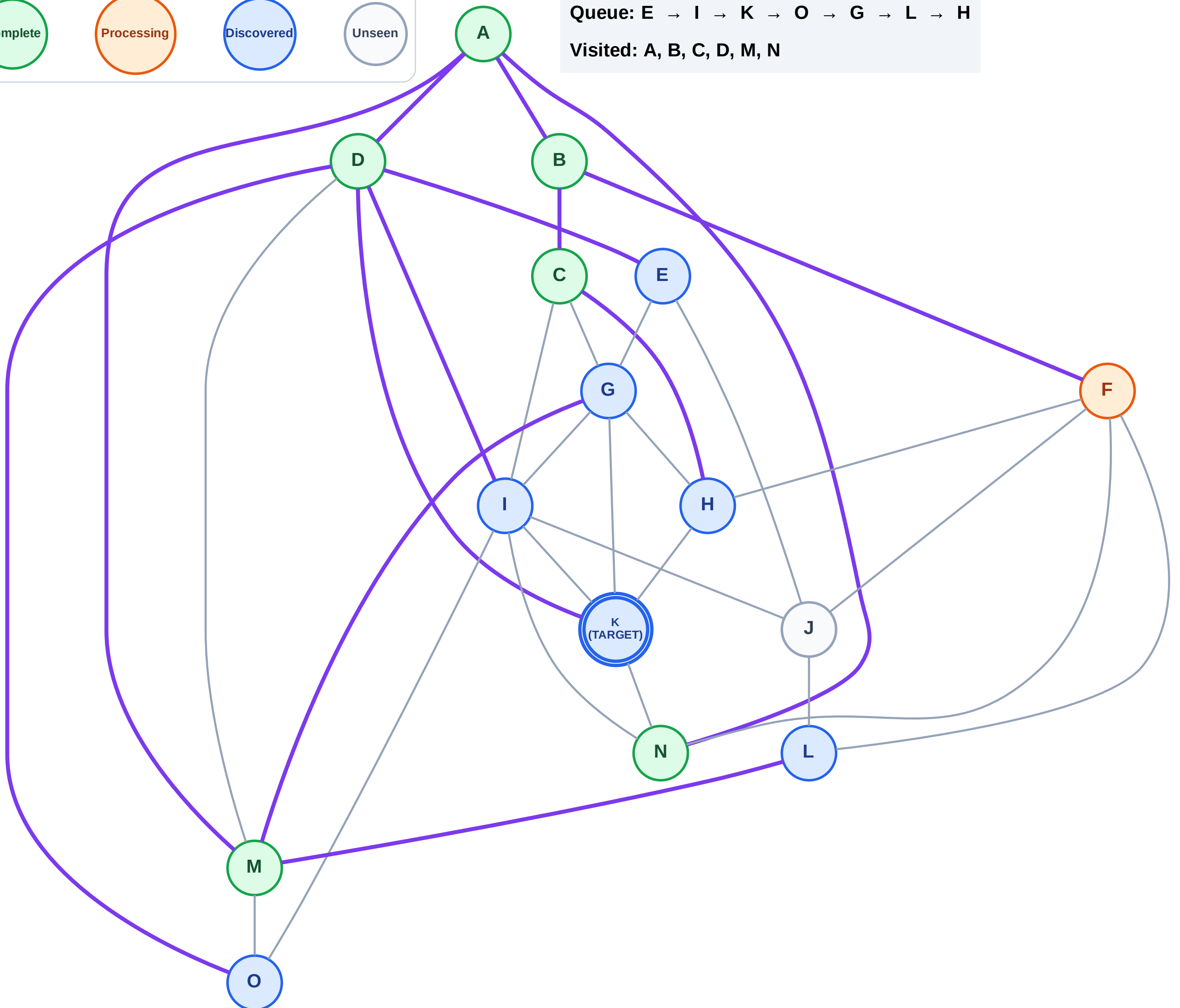
Step 14: Process node F

Legend



Queue: E → I → K → O → G → L → H

Visited: A, B, C, D, M, N



BFS Traversal

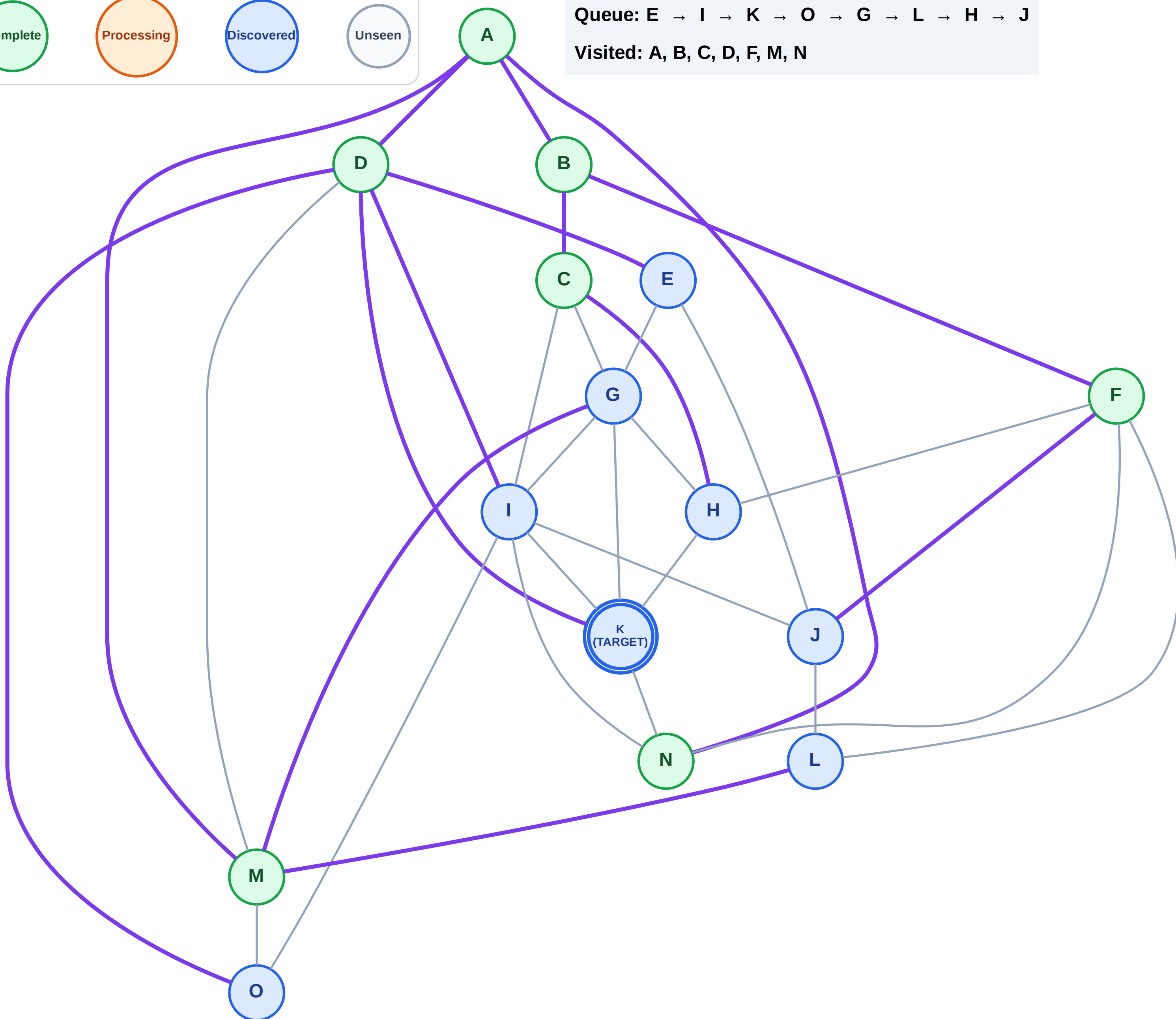
Step 15: Discover J from F

Legend



Queue: E → I → K → O → G → L → H → J

Visited: A, B, C, D, F, M, N



BFS Traversal

Step 16: Process node E

Legend



Complete



Processing



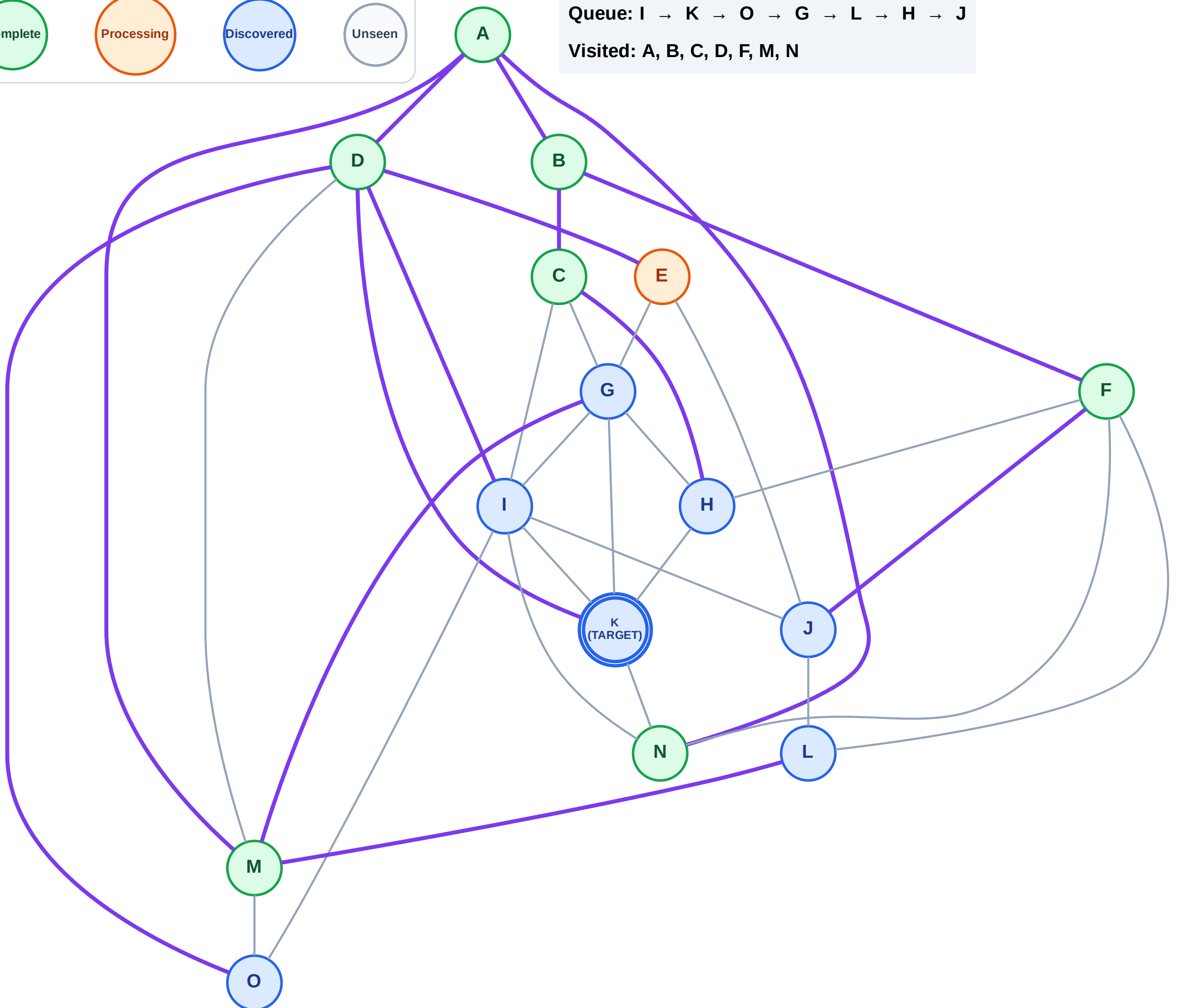
Discovered



Unseen

Queue: I → K → O → G → L → H → J

Visited: A, B, C, D, F, M, N



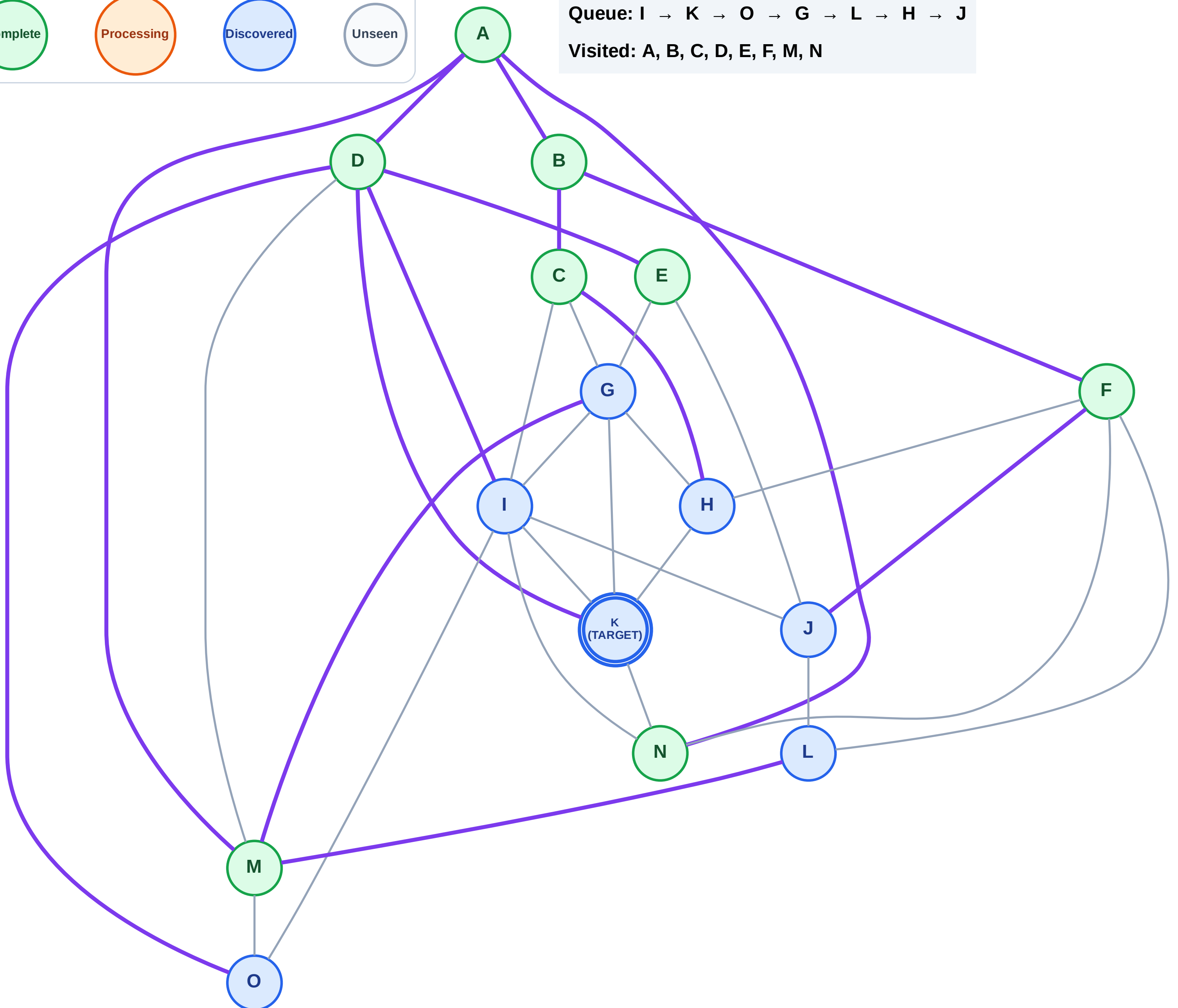
BFS Traversal

Step 17: No new neighbors from E

Legend



Queue: I → K → O → G → L → H → J
Visited: A, B, C, D, E, F, M, N



BFS Traversal

Step 18: Process node I

Legend

Complete

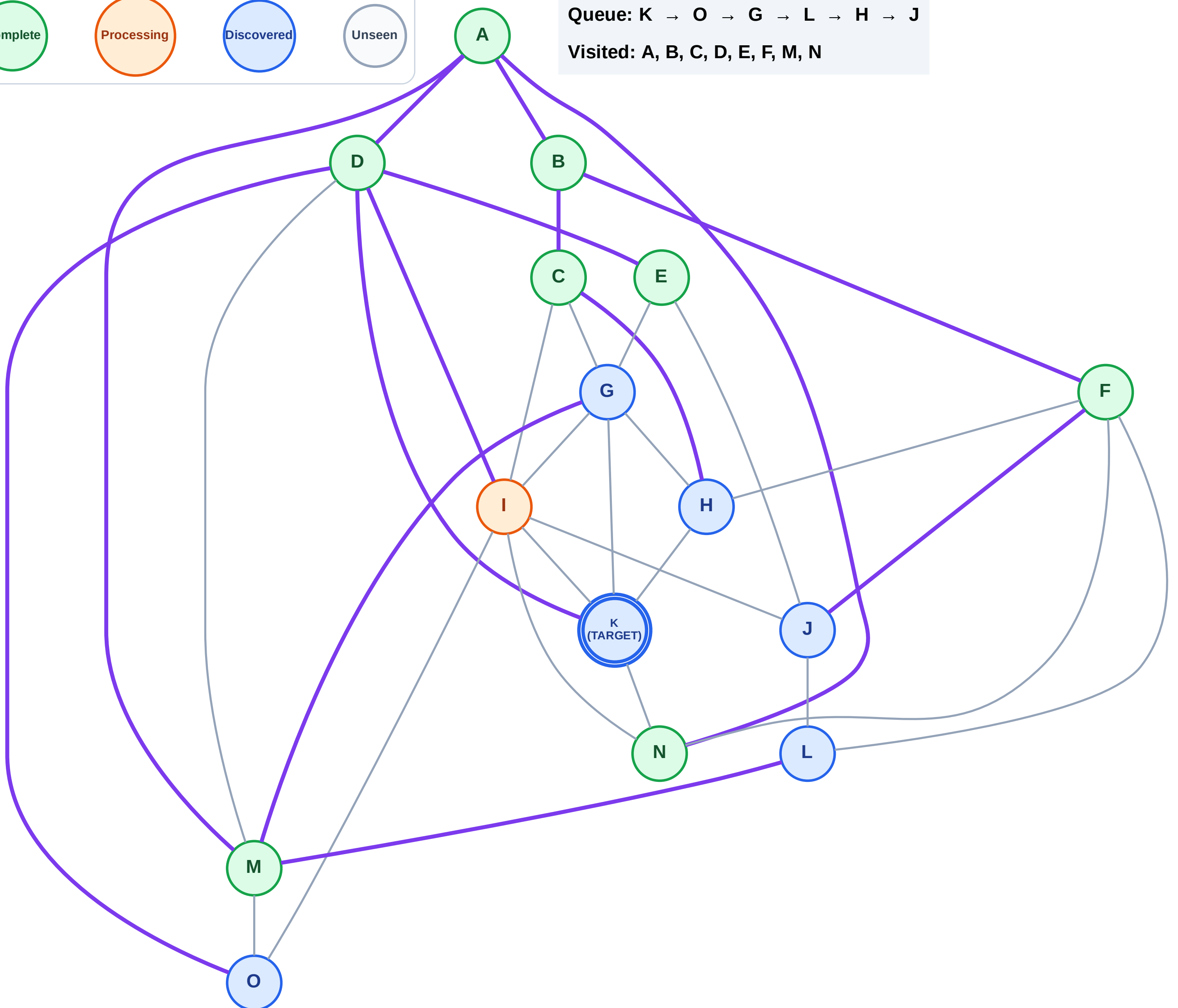
Processing

Discovered

Unseen

Queue: K → O → G → L → H → J

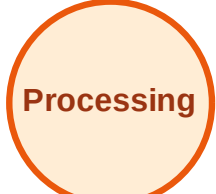
Visited: A, B, C, D, E, F, M, N



BFS Traversal

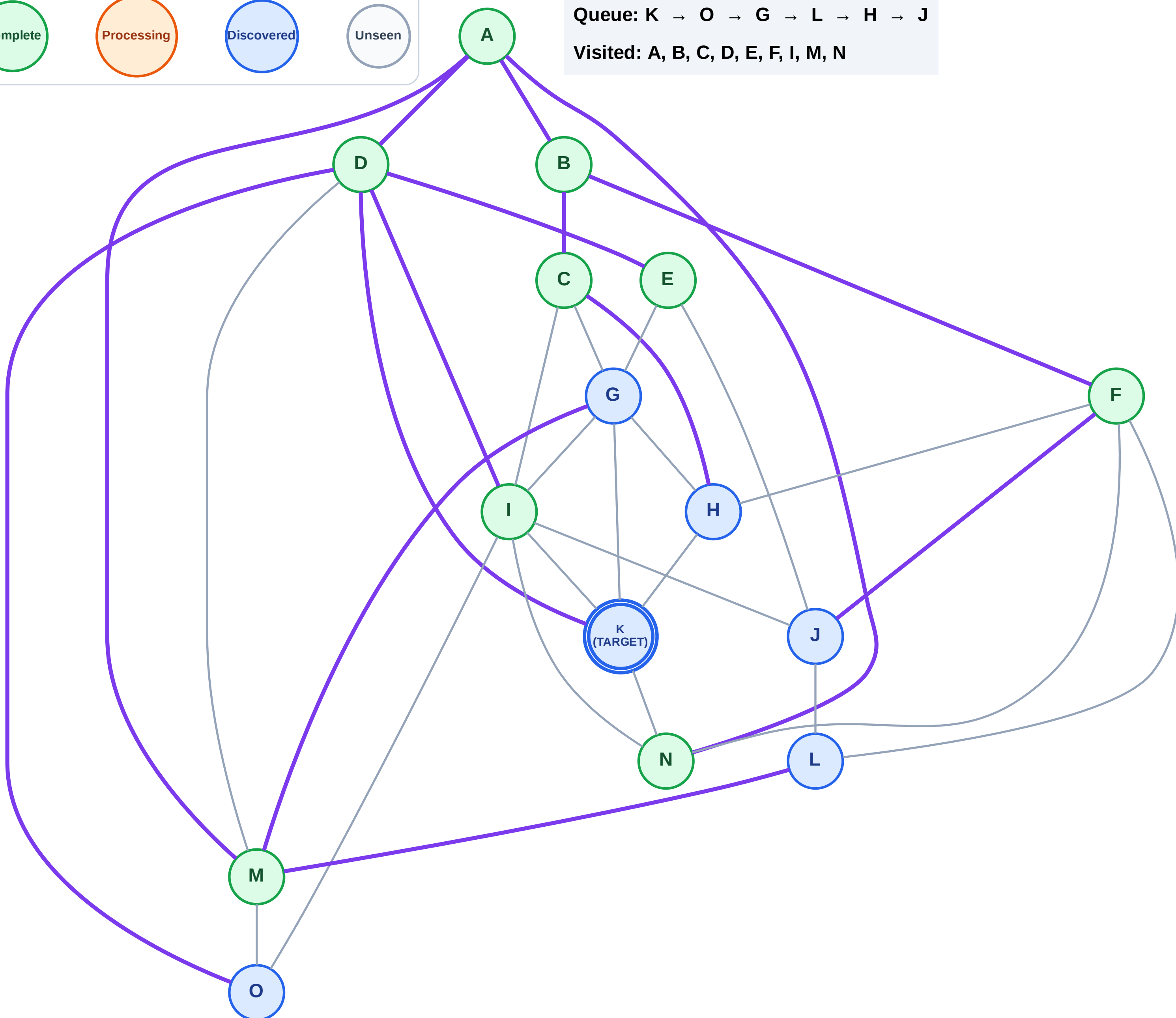
Step 19: No new neighbors from I

Legend



Queue: K → O → G → L → H → J

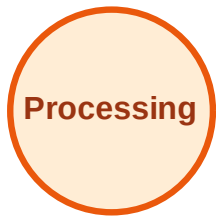
Visited: A, B, C, D, E, F, I, M, N



BFS Traversal

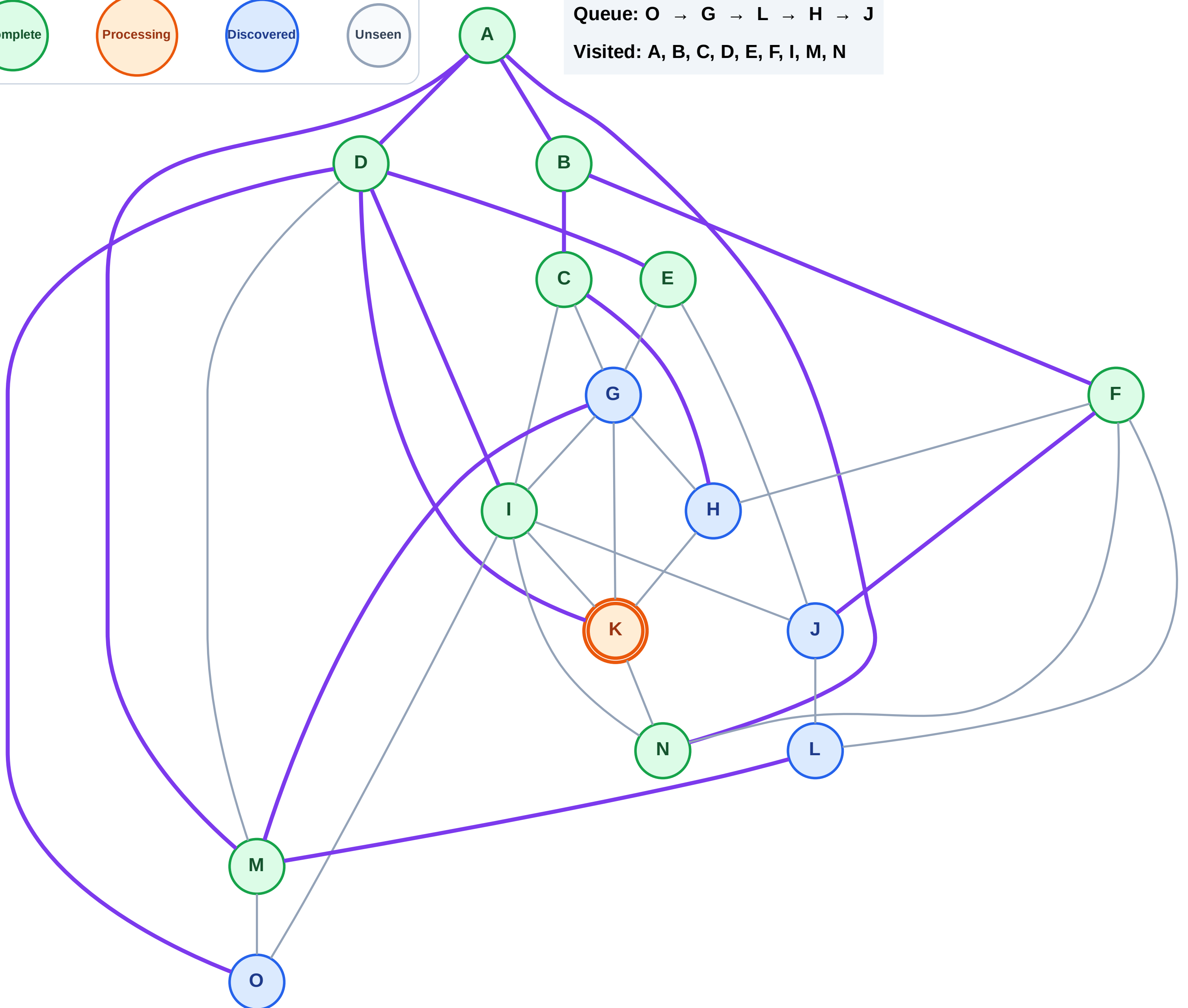
Step 20: Process node K

Legend



Queue: O → G → L → H → J

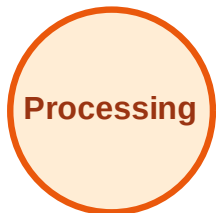
Visited: A, B, C, D, E, F, I, M, N



BFS Traversal

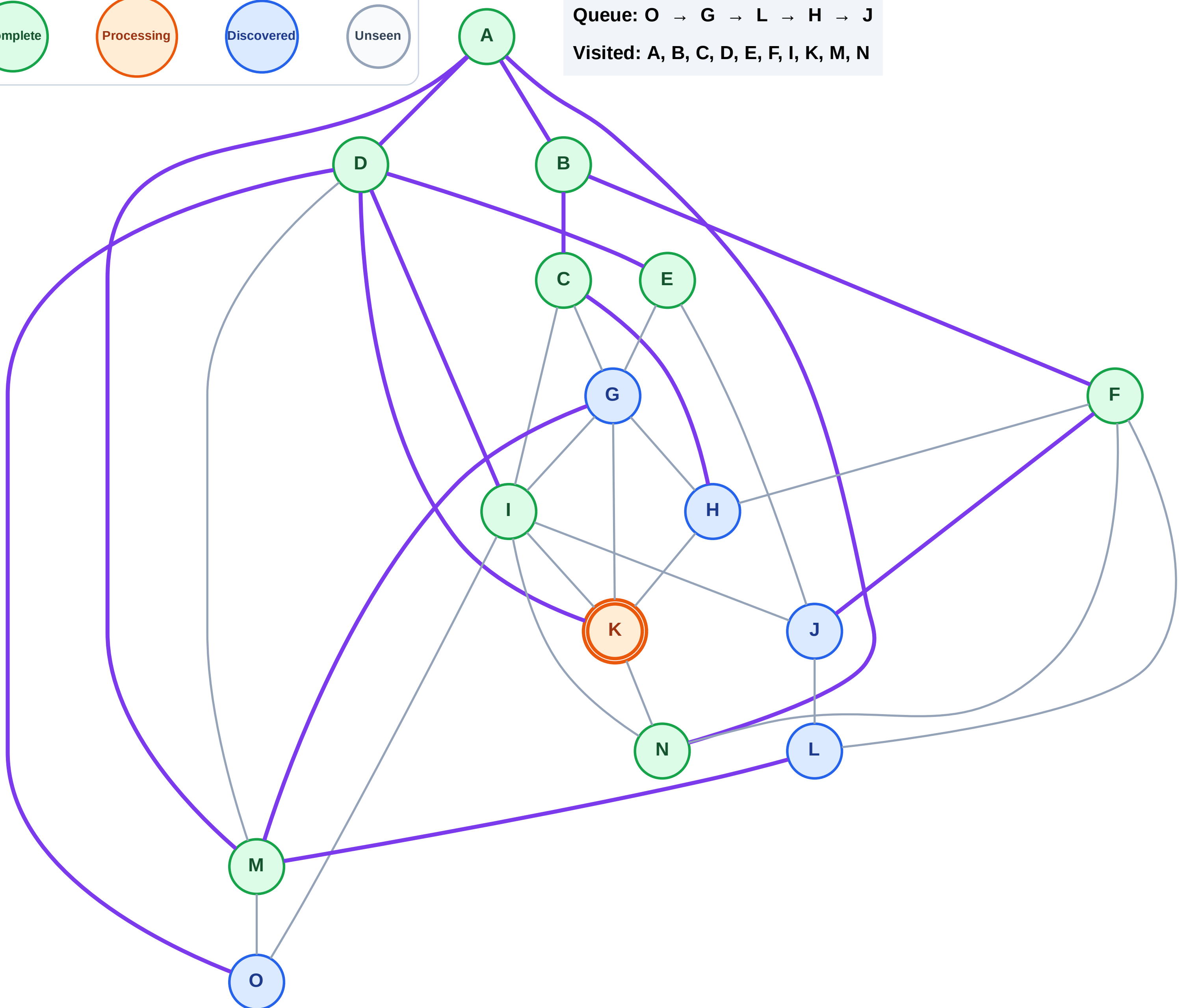
Step 21: TARGET NODE K REACHED! Path (2 hops): A → D → K

Legend



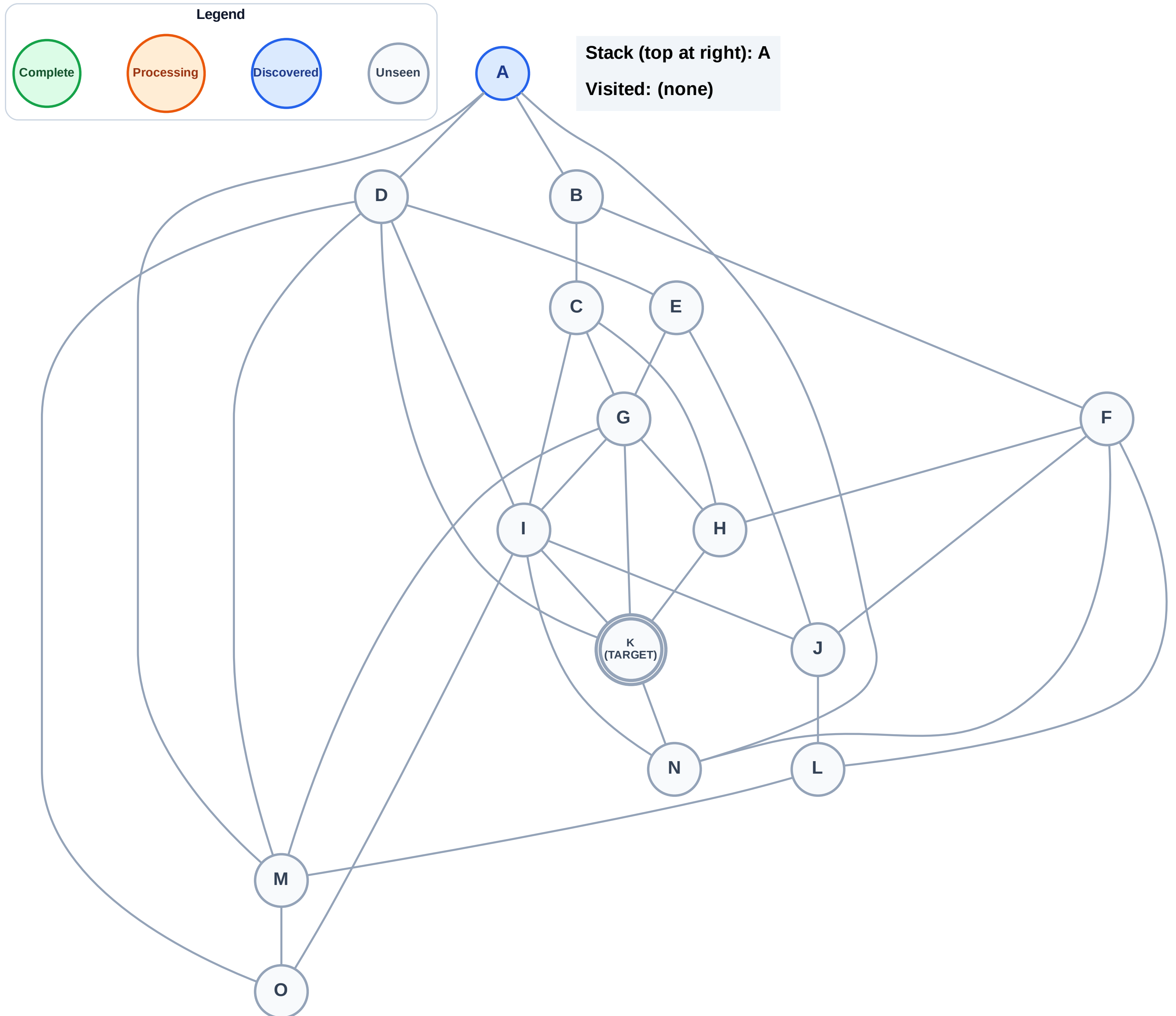
Queue: O → G → L → H → J

Visited: A, B, C, D, E, F, I, K, M, N



Step 1: Start at A (Seeking target node K)

Step 1: Start at A (Seeking target node K)



DFS Traversal

Step 2: Process node A

Legend

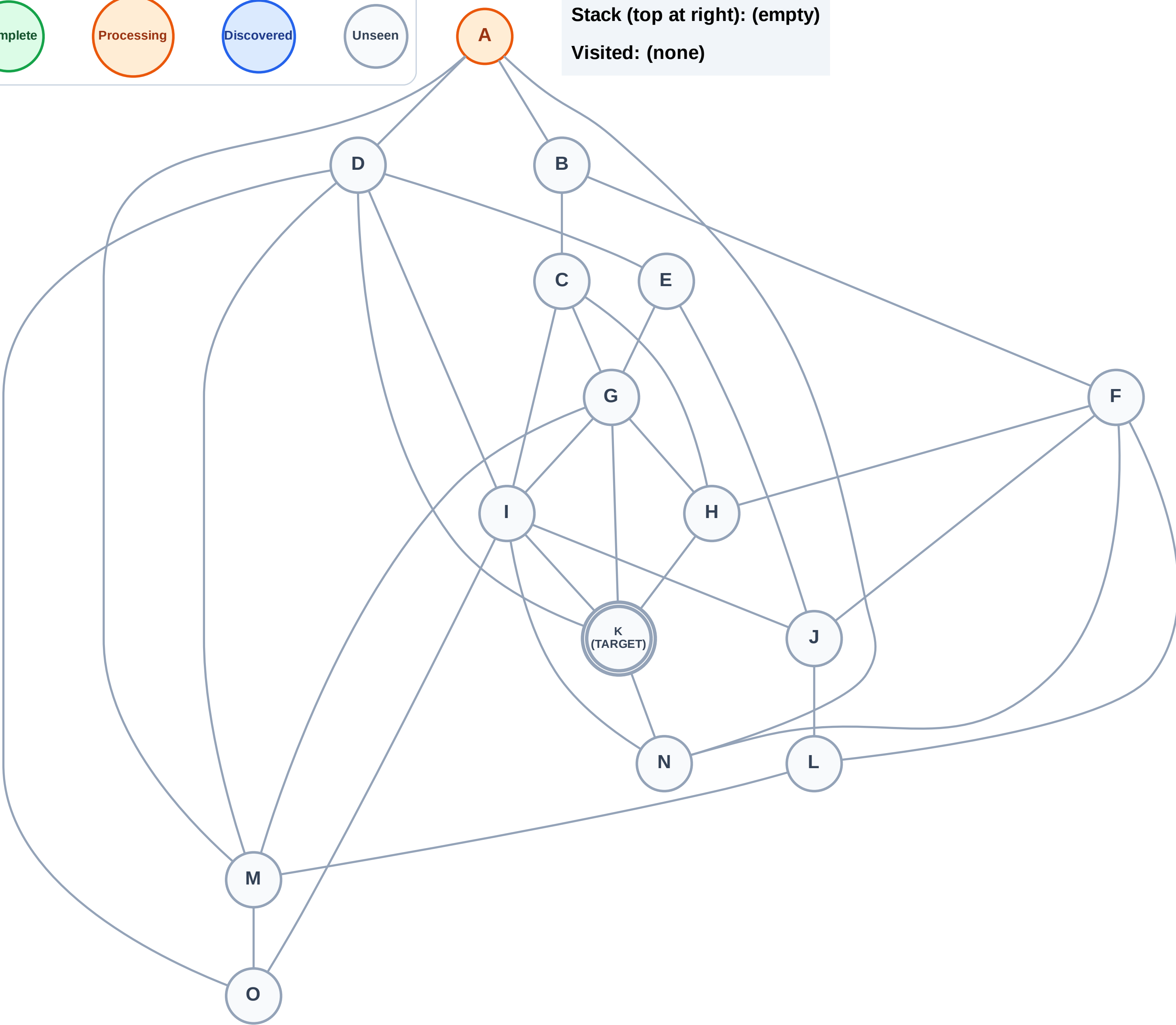
Complete

Processing

Discovered

Unseen

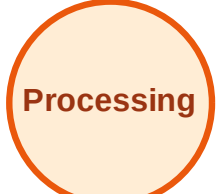
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

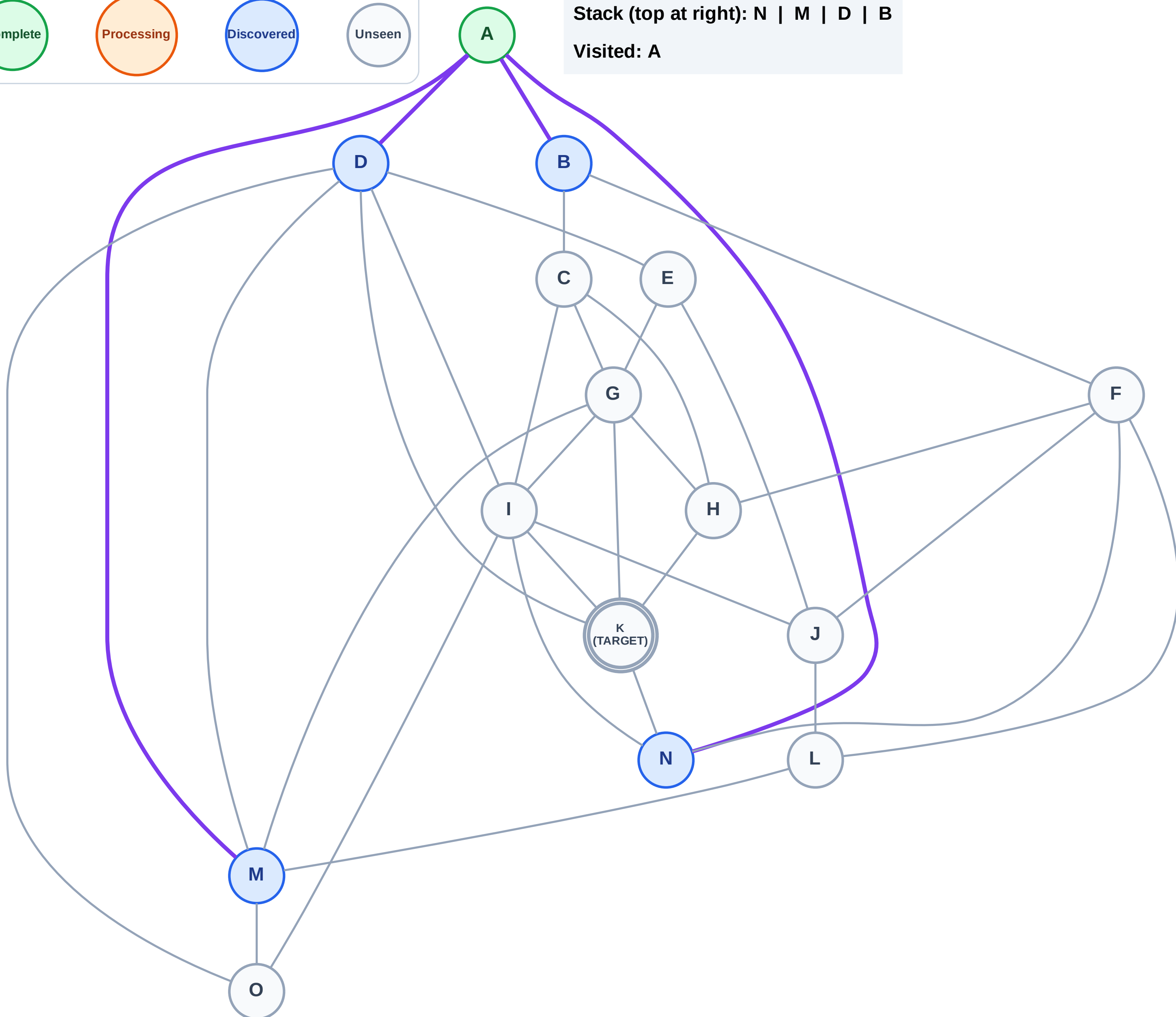
Step 3: Discover N, M, D, B from A

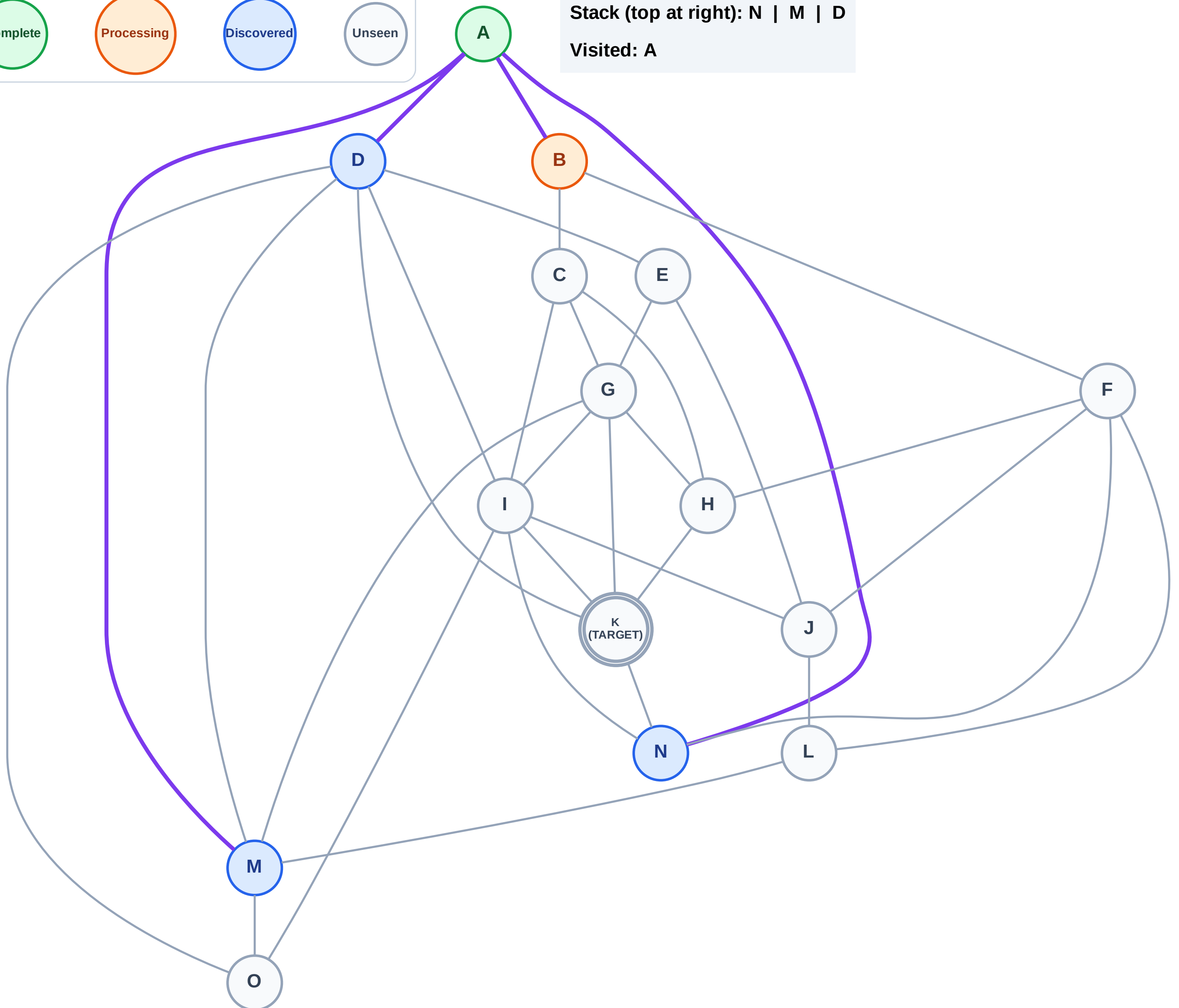
Legend



Stack (top at right): N | M | D | B

Visited: A





DFS Traversal

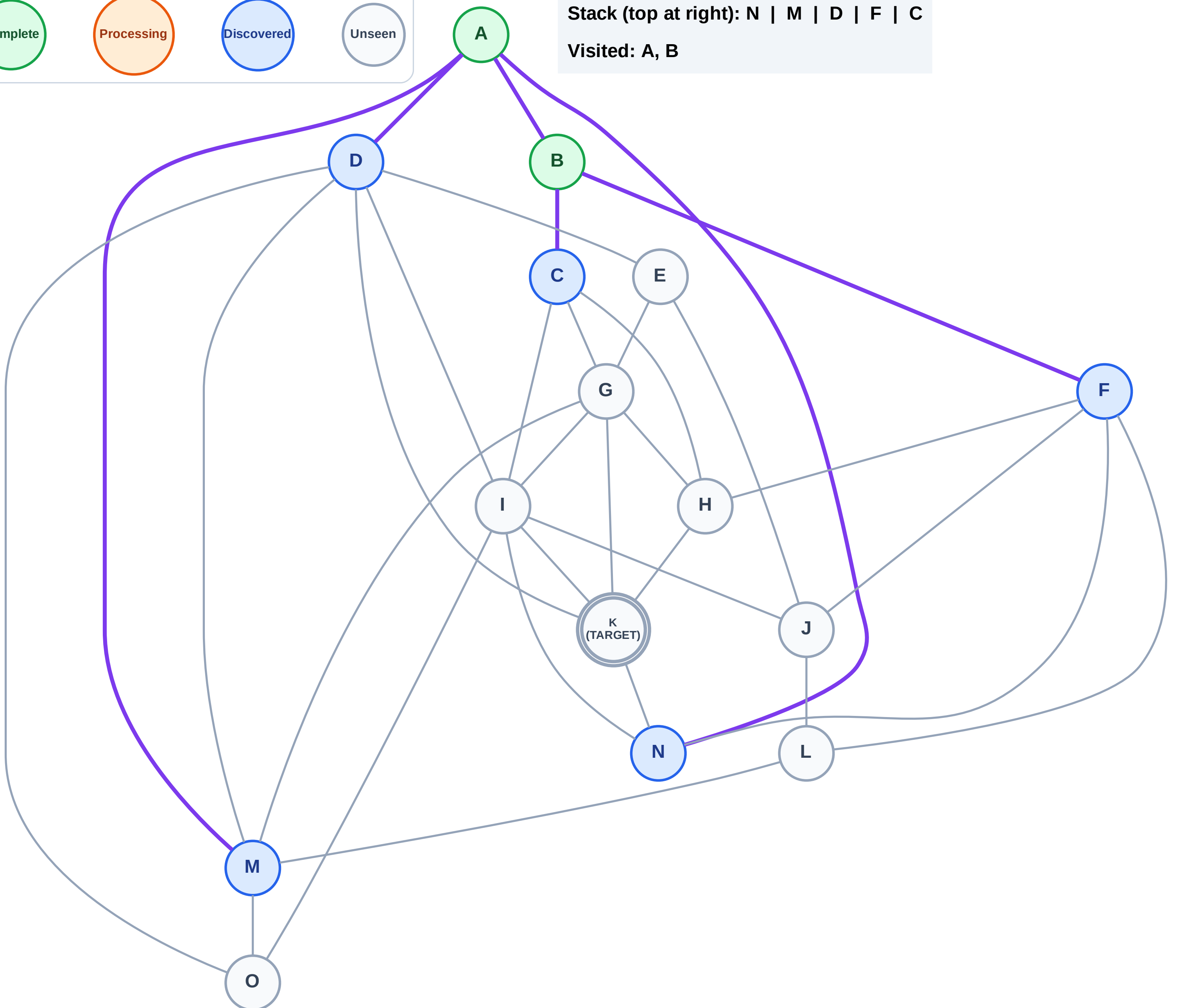
Step 5: Discover F, C from B

Legend



Stack (top at right): N | M | D | F | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

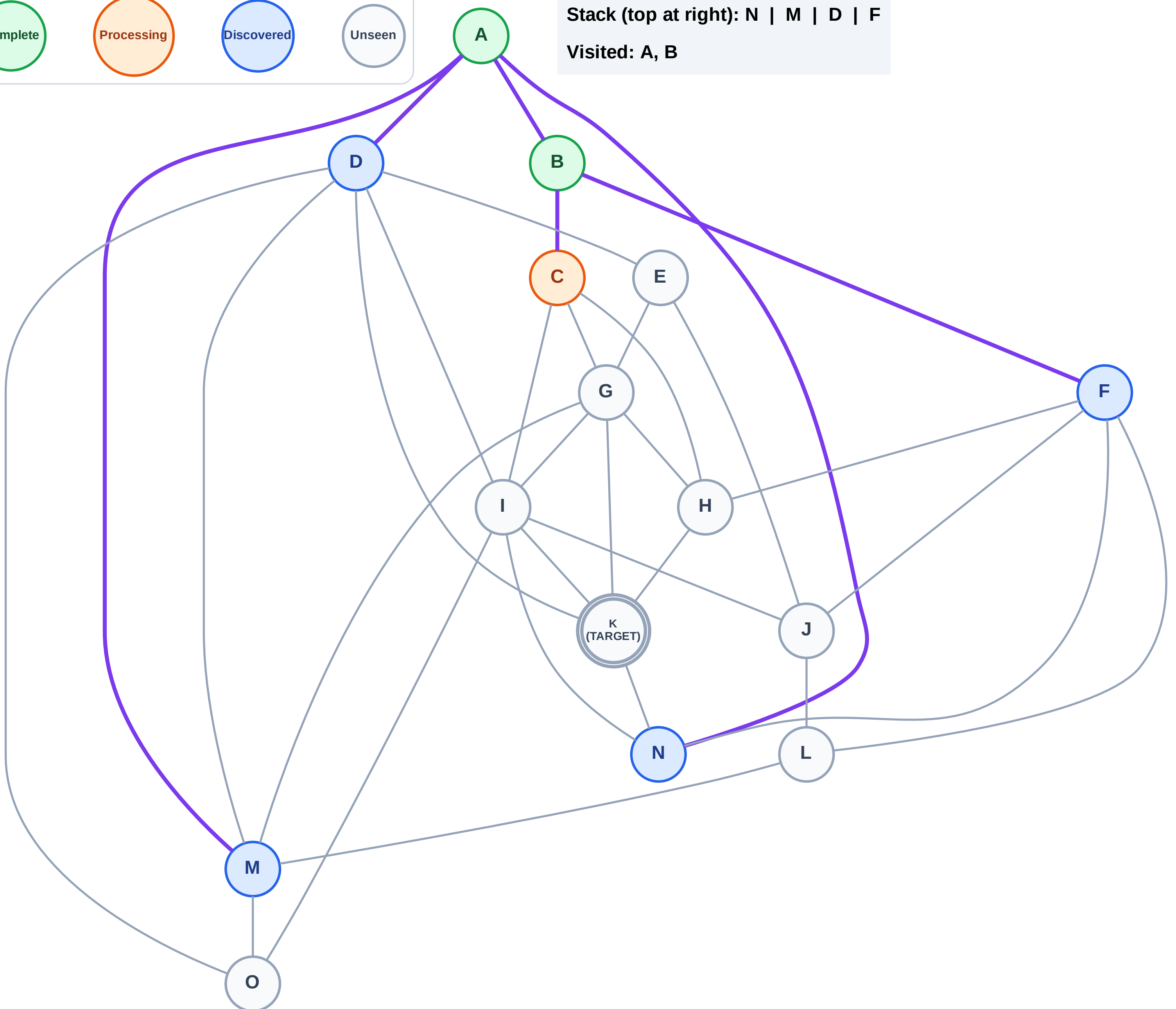
Processing

Discovered

Unseen

Stack (top at right): N | M | D | F

Visited: A, B



DFS Traversal

Step 7: Discover I, H, G from C

Legend

Complete

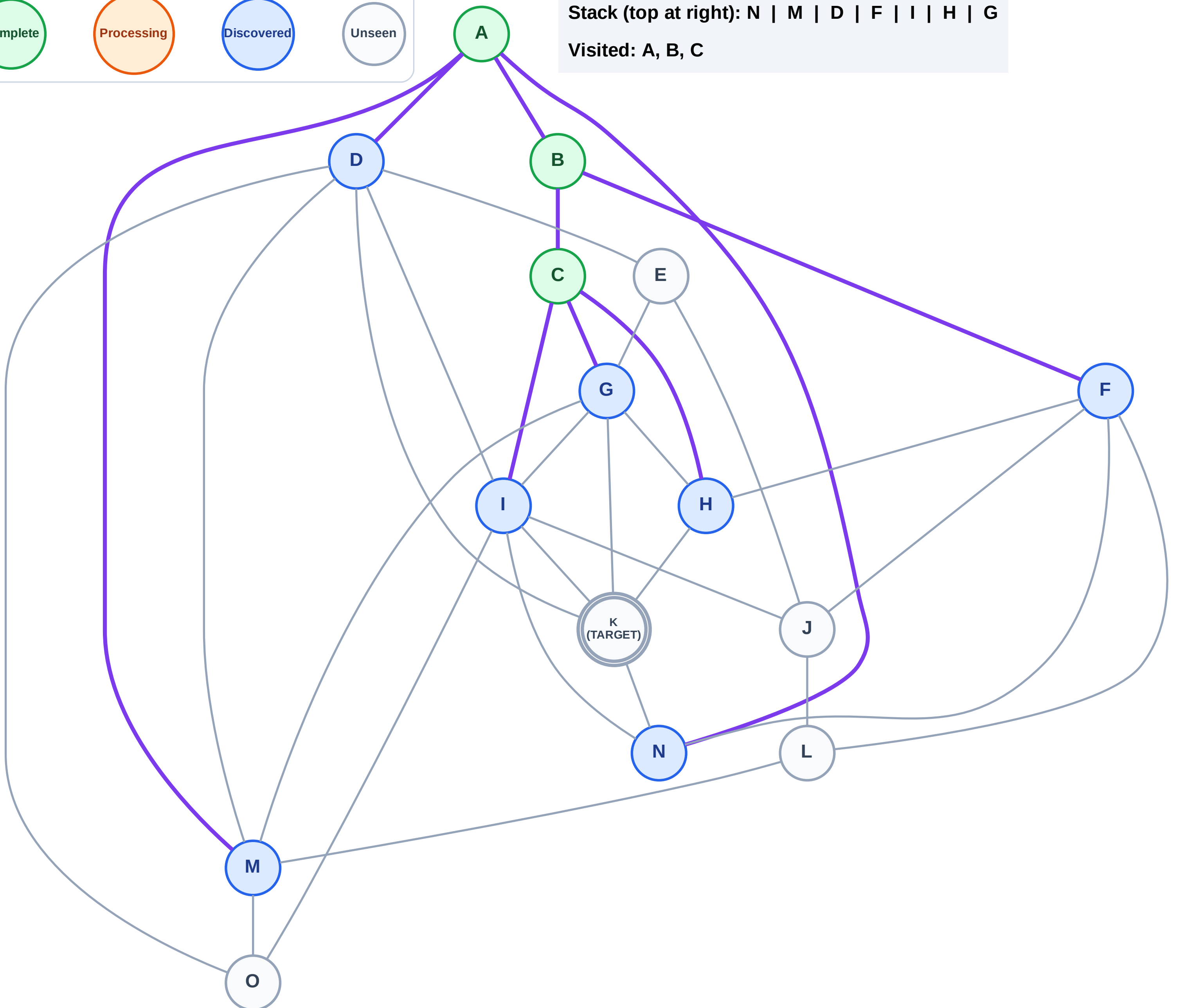
Processing

Discovered

Unseen

Stack (top at right): N | M | D | F | I | H | G

Visited: A, B, C



DFS Traversal

Step 8: Process node G

Legend

Complete

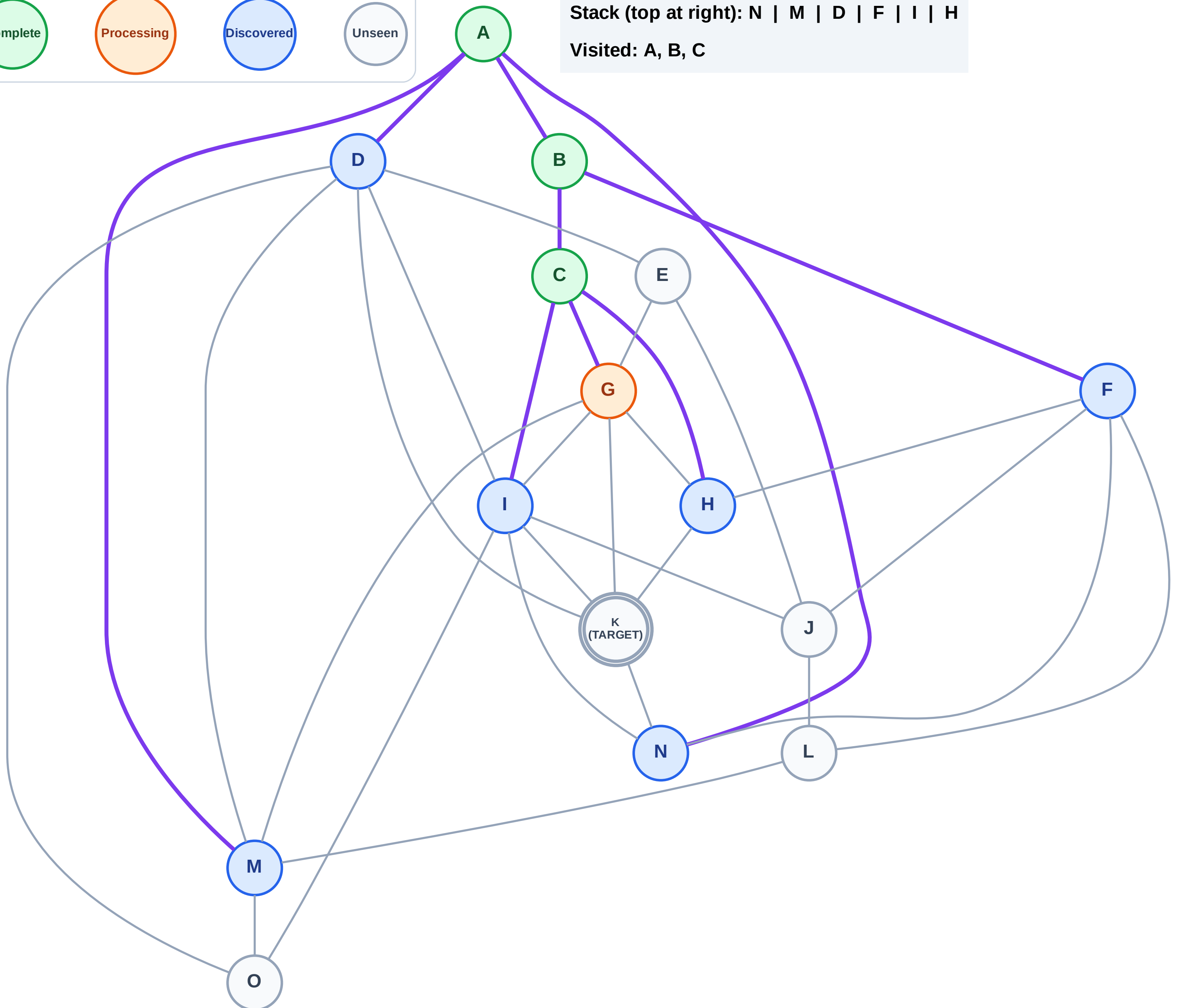
Processing

Discovered

Unseen

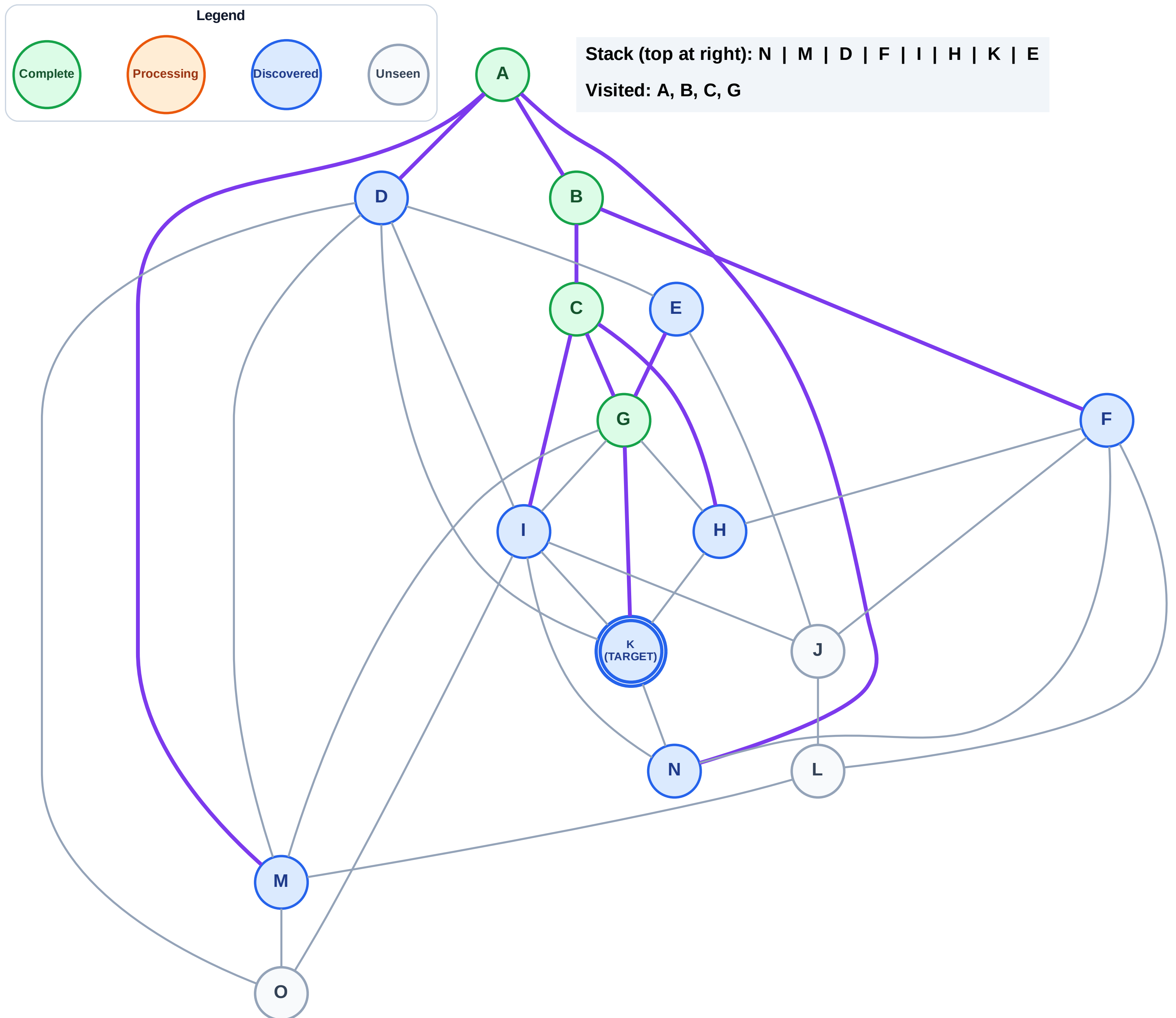
Stack (top at right): N | M | D | F | I | H

Visited: A, B, C



Step 9: Discover K, E from G

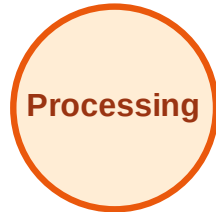
Step 9: Discover K, E from G



DFS Traversal

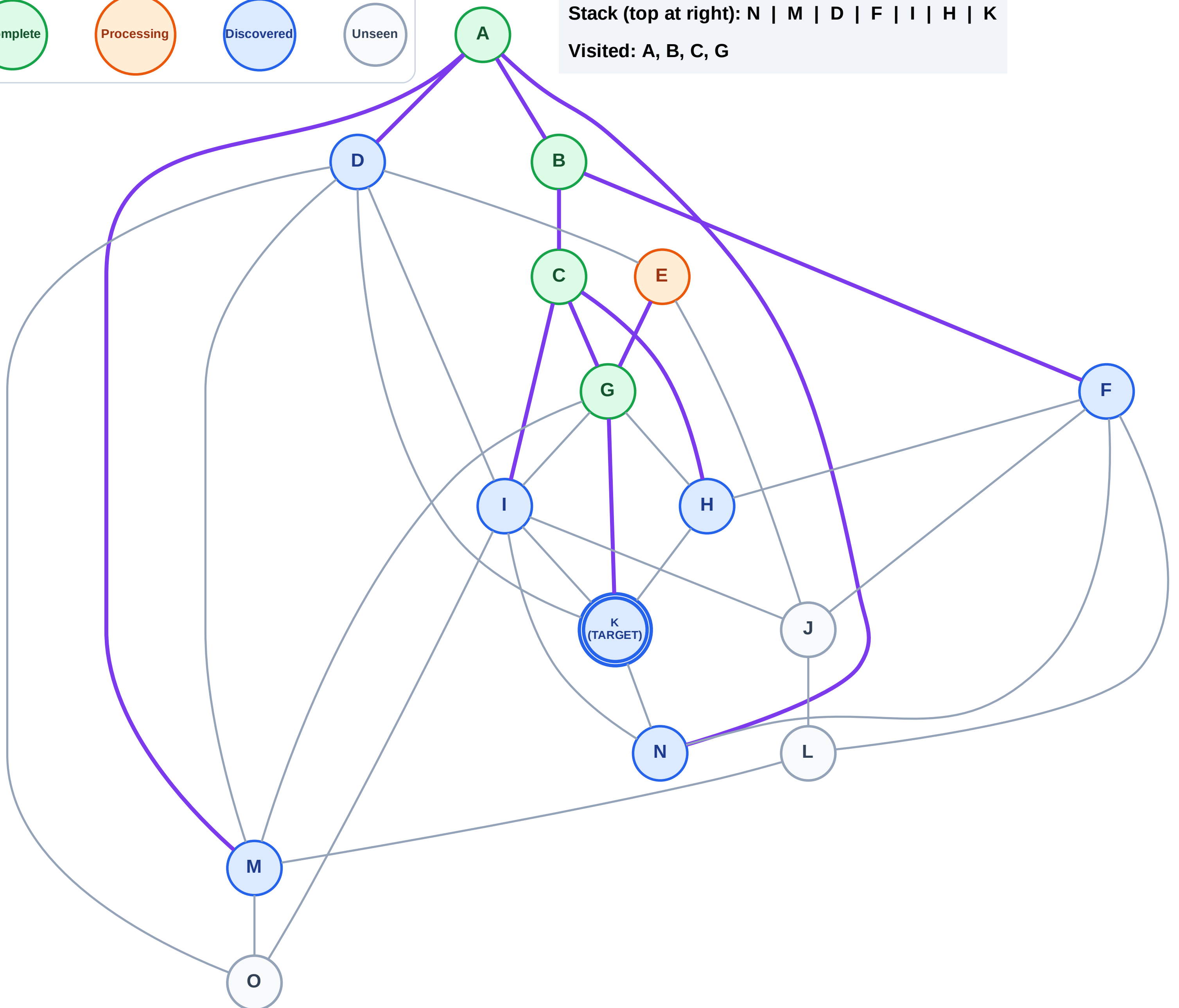
Step 10: Process node E

Legend



Stack (top at right): N | M | D | F | I | H | K

Visited: A, B, C, G



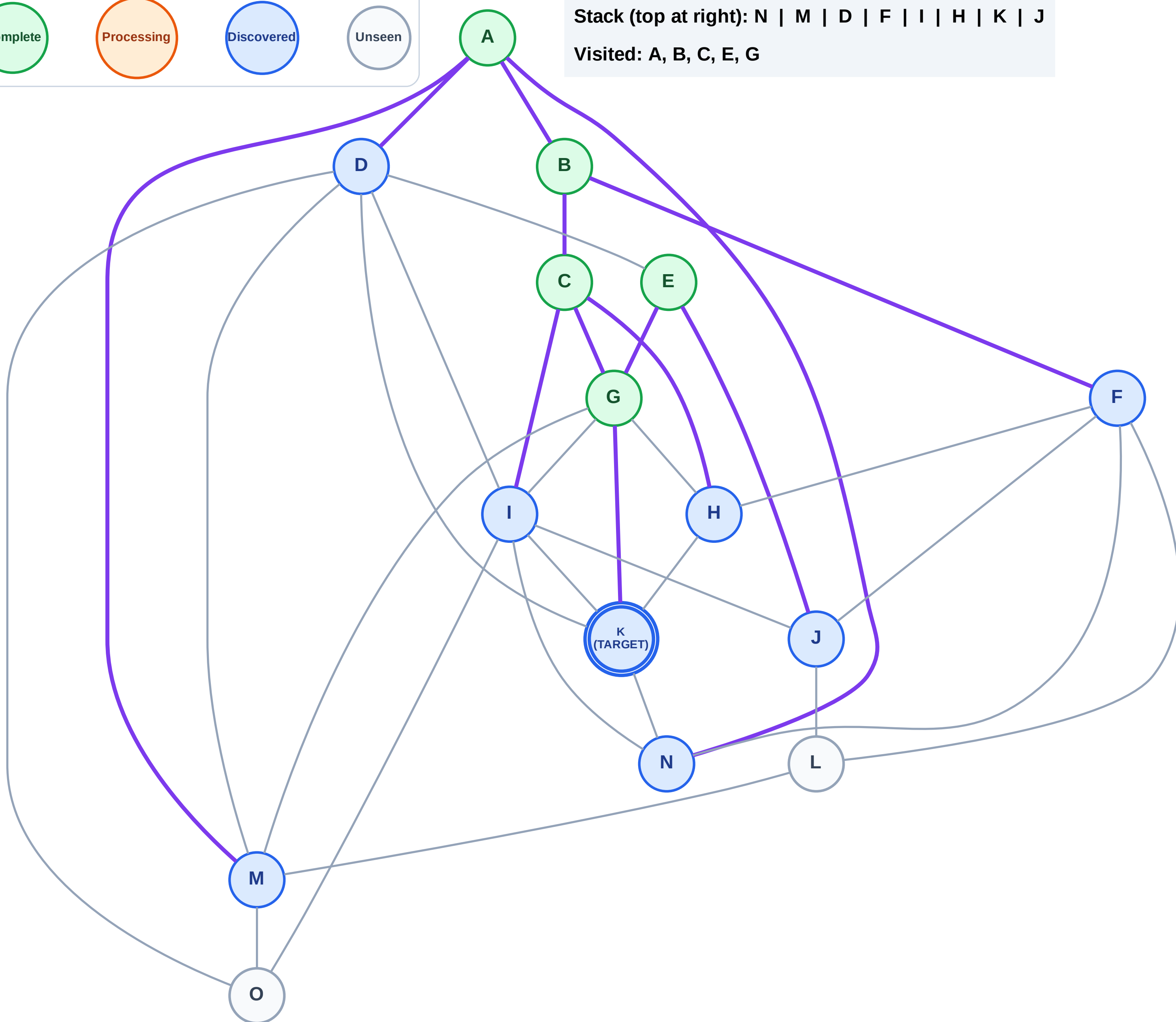
Step 11: Discover J from E

Stack (top at right): N | M | D | F | I | H | K | J

Legend

Processing

Unseen



DFS Traversal

Step 12: Process node J

Legend

Complete

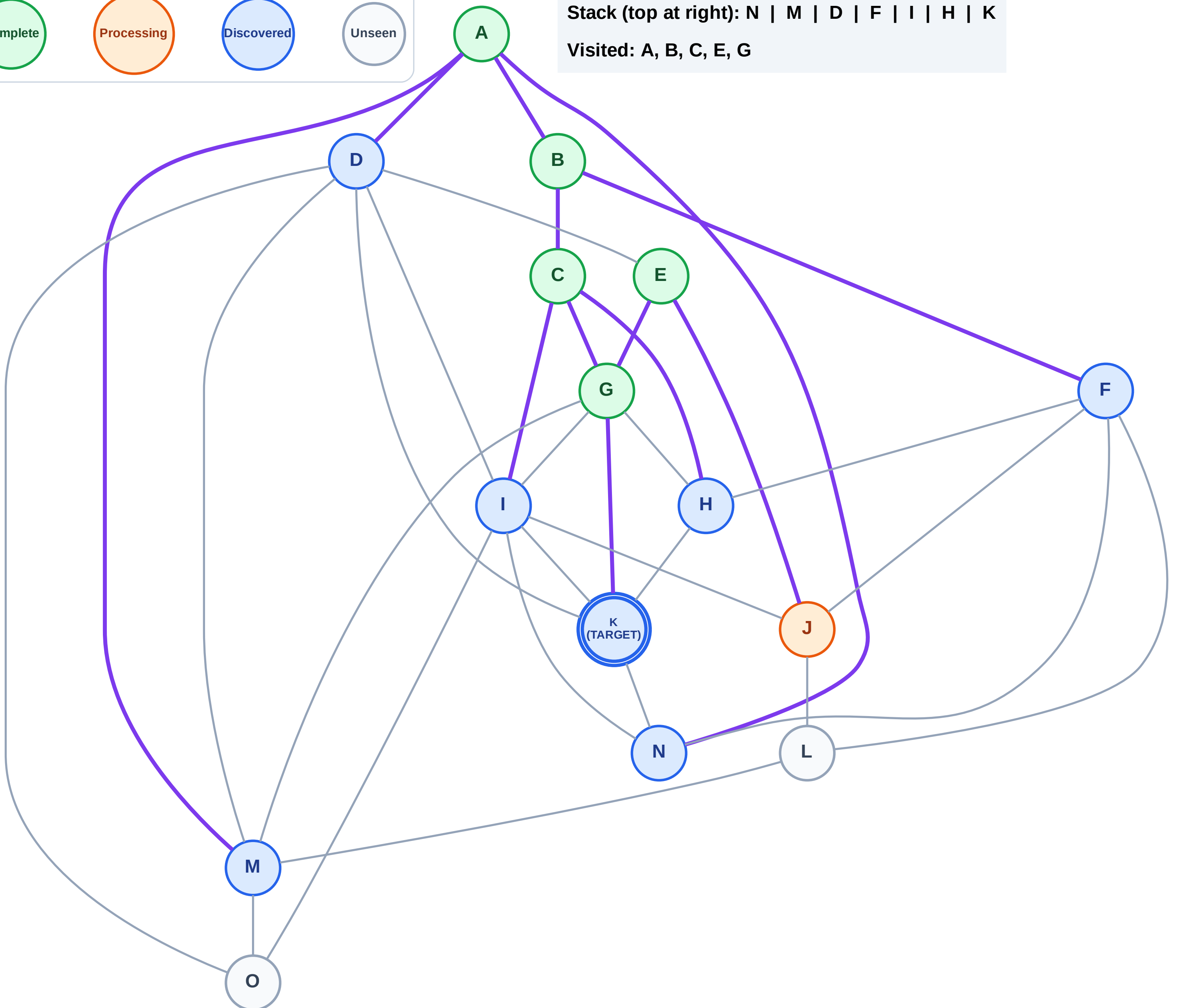
Processing

Discovered

Unseen

Stack (top at right): N | M | D | F | I | H | K

Visited: A, B, C, E, G



DFS Traversal

Step 13: Discover L from J

Legend

Complete

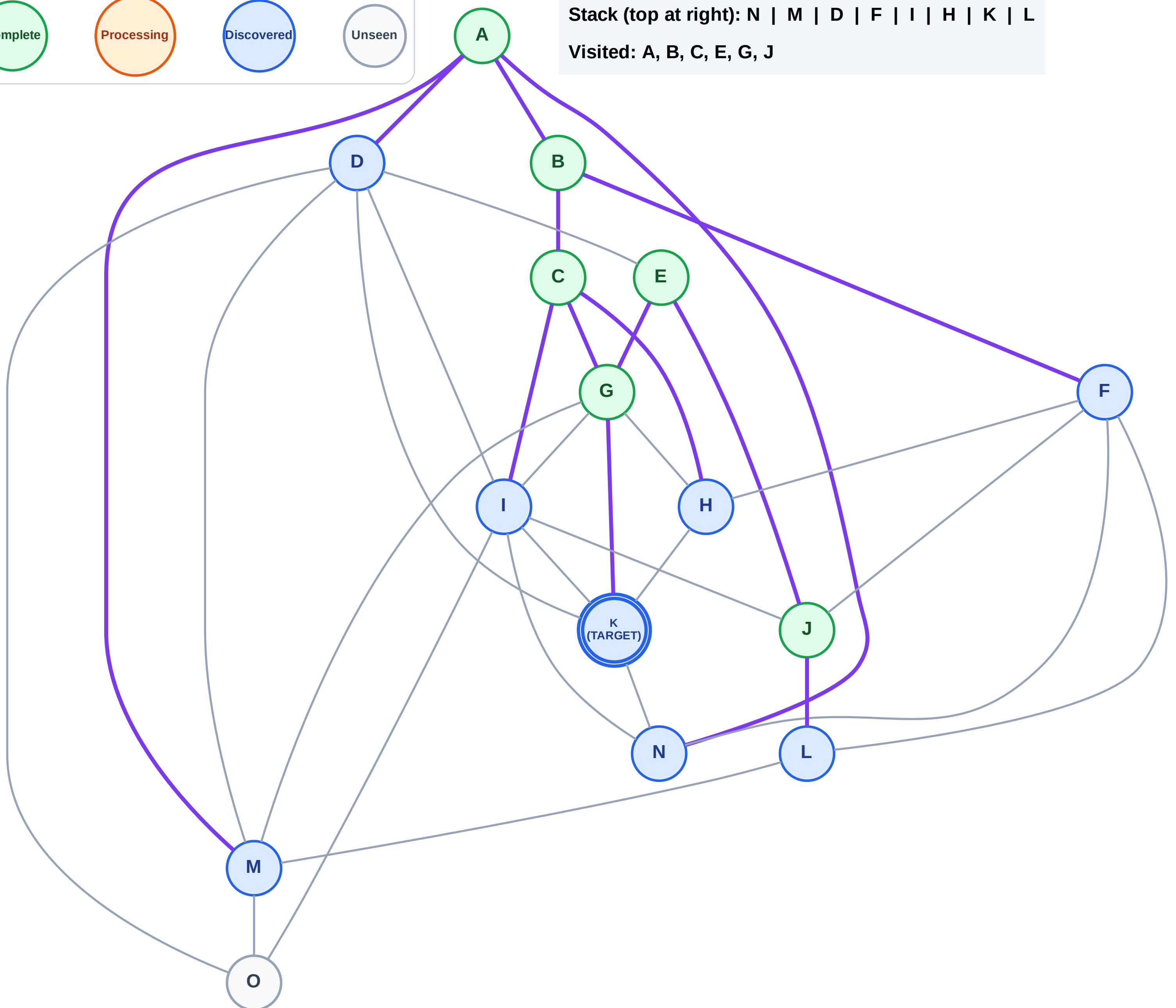
Processing

Discovered

Unseen

Stack (top at right): N | M | D | F | I | H | K | L

Visited: A, B, C, E, G, J



DFS Traversal

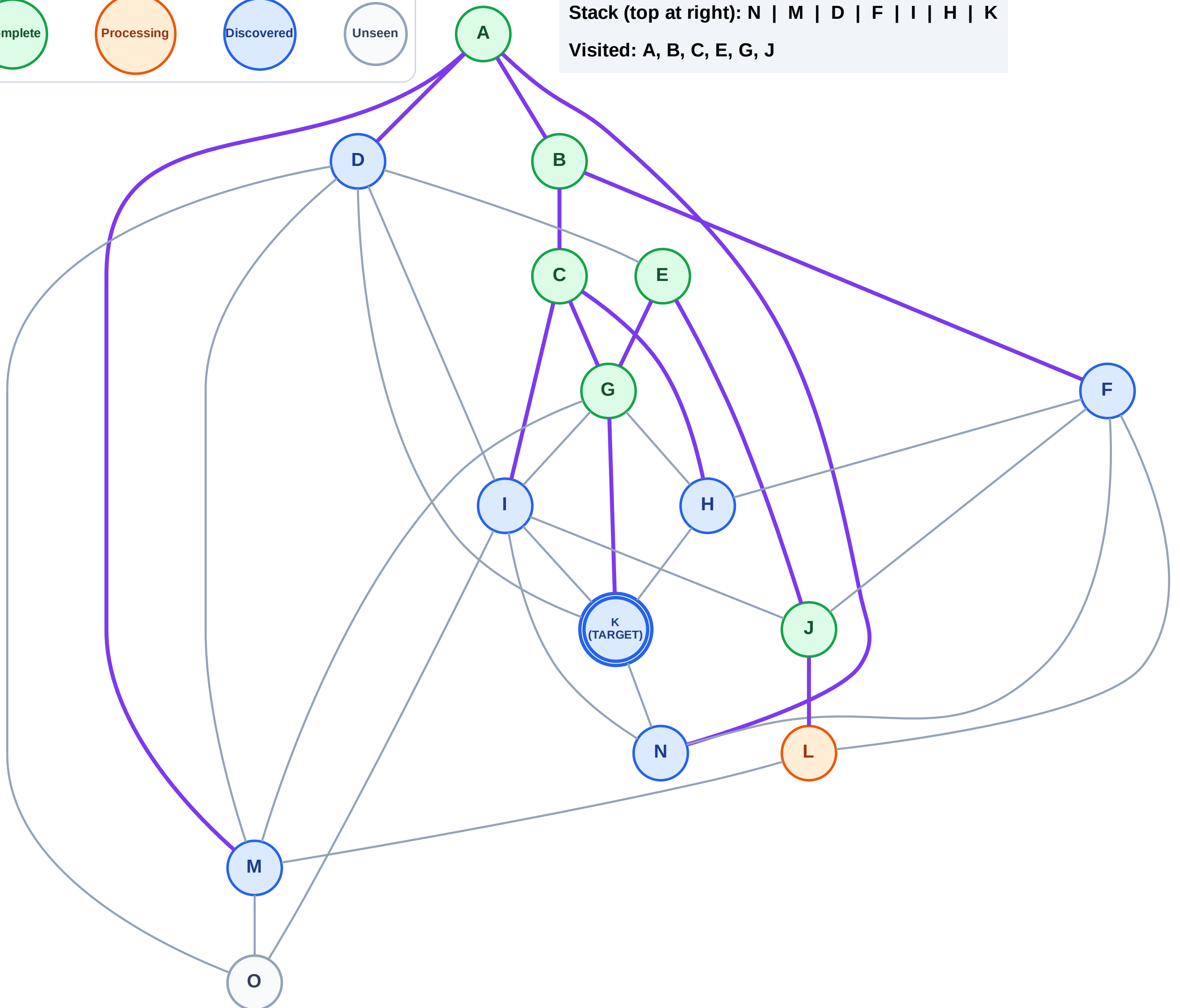
Step 14: Process node L

Legend



Stack (top at right): N | M | D | F | I | H | K

Visited: A, B, C, E, G, J



DFS Traversal

Step 15: No new neighbors from L

Legend

Complete

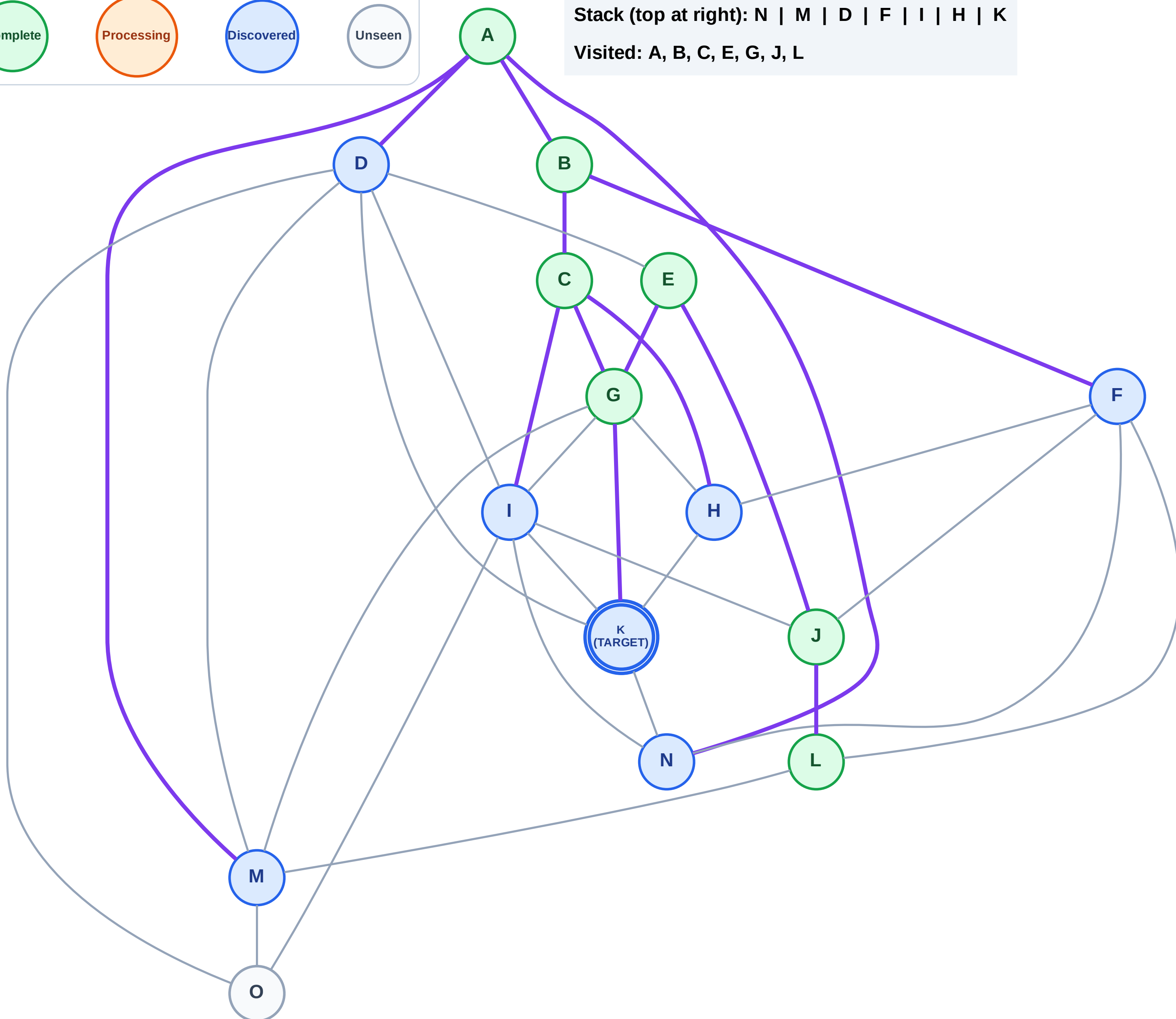
Processing

Discovered

Unseen

Stack (top at right): N | M | D | F | I | H | K

Visited: A, B, C, E, G, J, L



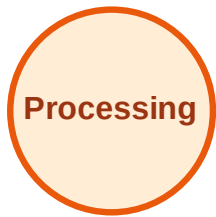
DFS Traversal

Step 16: Process node K

Legend



Complete



Processing



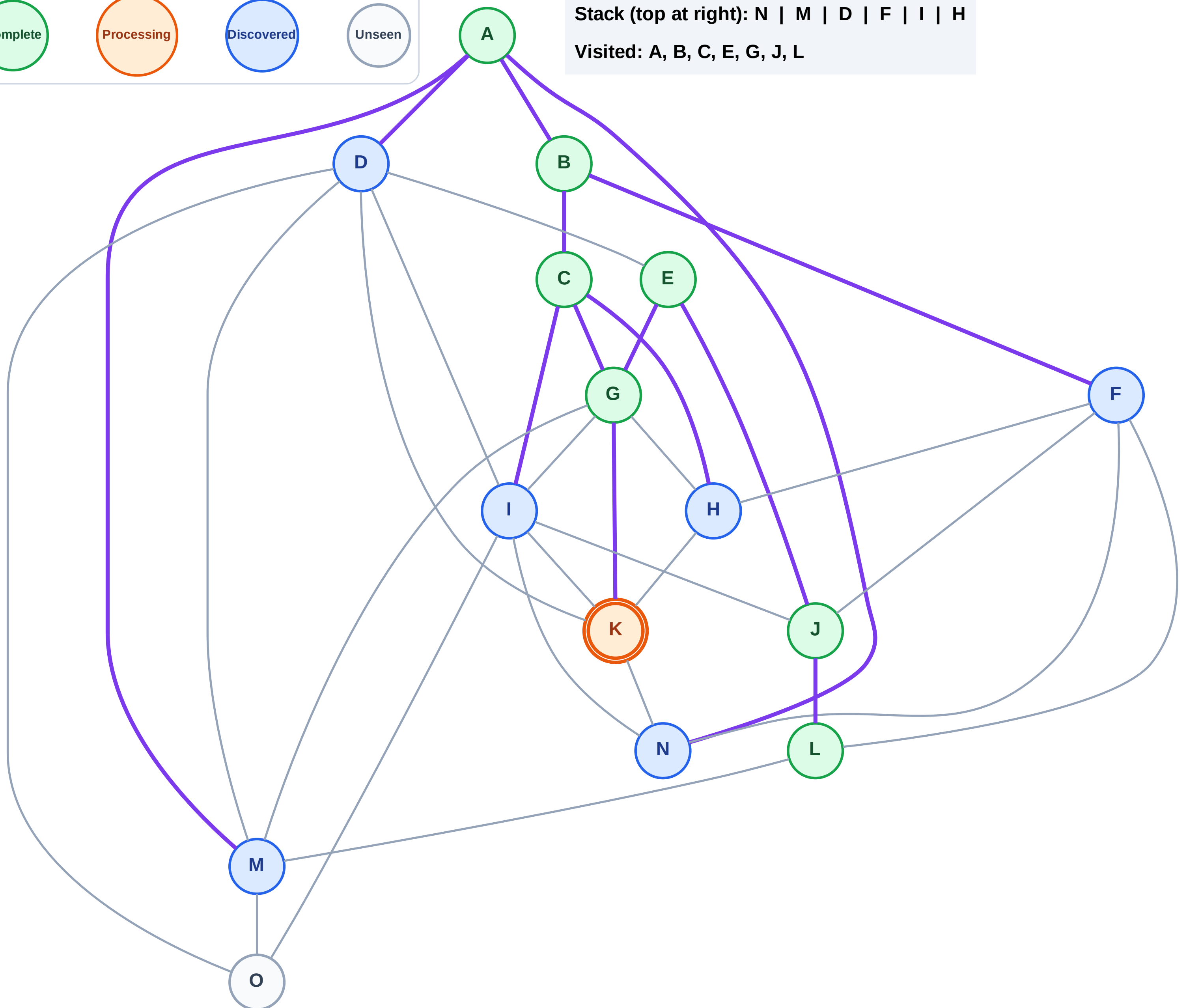
Discovered



Unseen

Stack (top at right): N | M | D | F | I | H

Visited: A, B, C, E, G, J, L



DFS Traversal

Step 17: TARGET NODE K REACHED! Path (4 hops): A → B → C → G → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): N | M | D | F | I | H
Visited: A, B, C, E, G, J, K, L

